

*Liquidity Provision under Flat Commissions in a Maker-Taker World**

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Abstract

Many equity exchanges charge the two parties of a trade different fees, often subsidizing liquidity providers. We examine this asymmetric, or maker-taker, pricing in a model where some investors pay fees directly to the exchange, while others pay them only on average, through flat commissions. When the latter investor group is large, maker-taker pricing is distortionary: incentivized by the lower liquidity provider fees (or rebates), investors who pay fees directly become de facto market makers. The aggregate gains from trade decline, and there is redistribution from flat-fee investors to those who pay fees directly. Reducing the fee asymmetry or lowering the share of investors who face flat commissions mitigate the distortions.

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The democratization of liquidity provision in modern limit order book markets allows all investors to propose the terms of a trade. This evolution remains a double-edge sword, as no assigned participant ensures consistently available liquidity—in good and bad times. To address this issue in equity markets, many exchanges offer cash incentives for filled limit orders. Exchanges that pay cash rebates to limit order traders that “make” liquidity typically levy higher fees to remove or “take” liquidity through market orders.

Referred to as “maker-taker” pricing,¹ the practice remains an issue in regulatory and policy debates on market structure: in the U.S., the Securities and Exchange Commission proposed a controversial transaction fee pilot that places caps on maker rebates to study, in part, their impact market and execution quality.² Several exchanges have raised opposition to the pilot, maintaining that market participants benefit from the increase in competition for liquidity provision spurred by the rebates. Proponents of the study—including the Investor’s Exchange as one of the few exchanges in support—argue, however, that such competition lowers the execution probability of long-term investors’ limit orders, which compels them to trade with more expensive market orders.³

Industry debates notwithstanding, the academic literature questions the economic impact of maker-taker pricing on equity markets. Angel, Harris, and Spatt (2011) and Colliard and Foucault (2012) argue that maker-taker fees affect trading only through the net fee per trade remitted to the exchange, but that the split of this fee into a maker rebate and a taker fee—absent regulatory or market frictions—is irrelevant. But, when liquidity provision incentives do not manifest identically in the fee schedules of all investors, how does this affect the quality of liquidity, and investor welfare?

¹IOSCO Consultation Report, International Organization of Securities Commissions (2011), p. 19.

²See Securities and Exchange Commission (14-03-2018). Note: the pilot was shelved in June of 2020 after a D.C. Circuit Appellate Court ruled that the SEC had overstepped its authority in imposing the new rules. (<https://tax.thomsonreuters.com/news/appeals-court-tosses-secs-maker-taker-pilot/>).

³See, comment letters from e.g., Citigroup Global Markets Inc; Council of Institutional Investors, and IEX, in favor of the pilot; NYSE and NASDAQ provide letters in dissent.(<https://www.sec.gov/comments/s7-05-18/s70518.htm>)

Our motivation is grounded in the reality of how maker-taker fees apply differently across investor groups: those who access the exchange directly may receive maker rebates for filled limit orders (and pay taker fees for market orders), but many investors access the exchange through a broker who pays fees and receives rebates on their behalf, often charging a flat fee per trade to their clients.⁴ The disparity in how maker-taker fees are paid by investors effectively creates a “two-tiered system” for liquidity provision, which we show plays a key role toward the impact of maker-taker pricing on market quality and investor welfare.

We build upon the model of Foucault (1999) to study the role that cash incentives for liquidity provision play in equity markets when investors face different fee schedules. Motivated by private values, investors enter the market sequentially to trade a single unit of a risky security. These investors trade using market or limit orders; if using a limit order, they must specify a price at which the order must be filled—the ‘limit price’. A subset of investors submit orders directly to the exchange, and pay maker-taker fees on a per-trade basis. The remainder trade through a broker that relays the orders to the exchange. The broker charges a flat fee per-trade that is independent of order type, and set equal to the average trading fees that the broker remits to the exchange.

Consistent with the previous literature (see Angel, Harris, and Spatt (2011) and Colliard and Foucault (2012)), when all investors face the same fee schedule—either maker-taker or flat-fee—the split does not play a role in our model: any increase in the maker rebate is passed to the takers through a narrower bid-ask spread, exactly offsetting an increase in the taker fee.⁵

In the two-tiered system, however, maker-taker pricing is distortionary. In equilibrium, investors choose the limit price of their limit orders to ‘target’ (a subset of)

⁴See, e.g., the Interactive Brokers: <https://www.interactivebrokers.com/en/index.php?f=commission>

⁵When discussing our results, we focus on the prevalent industry practice of a negative maker fee, or a rebate, but our analysis extends to the case of a positive maker fee.

investors that arrive in the subsequent period. Because the flat fee charged by the broker is an average of the maker and taker fees that the broker collects, the flat fee per market order is less than the taker fee paid by investors that access the exchange directly. As a result, “flat-fee investors” (those who pay flat fees to a broker) are willing to trade against limit orders with less-favorable limit prices than maker-taker investors (those who pay fees directly to the exchange). Hence, there are two sets of equilibrium buy and sell limit prices: more favorable prices that target both types of investors, and less favorable prices, that target only flat-fee investors.

Because flat-fee investors pay a higher fee (or receive a lower rebate) for submitting limit orders, they offer less favorable limit order prices than maker-taker investors. In contrast, maker-taker investors assume a more prominent role in liquidity provision, becoming de facto market makers: they post limit orders at narrower spreads and their limit orders execute with higher probability. Maker-taker investors are also less likely to use market orders versus flat-fee investors, preferring to submit limit orders when facing the less-favorable limit orders of flat-fee investors. The result is lower overall trading volume, and higher profits (on average) accruing to maker-taker investors, as they trade with limit orders more often, thereby receiving the rebate. While much of the literature focuses on market-making in the context of a speed advantage (See e.g., Menkveld (2013), Han, Khapko, and Kyle (2014), and Üslü (2019)), our model argues a complementary channel for liquidity provision incentives. Differences in fee schedules across market participants induce traders who pay access fees and receive liquidity rebates directly to become market makers *independent of a speed advantage*.

Beyond the direct impacts on liquidity provision, we employ our model to examine investor welfare by defining a measure that reflects allocative efficiency (see for example, Bessembinder, Hao, and Zheng (2015)). We find that social welfare co-moves with volume, and hence, a two-tiered system reduces welfare.

Crucially, the distortionary impact of the maker-taker pricing can be mitigated by reducing the flat-fee investor incentives for posting unfavorable, wide-spread limit orders. Our analysis suggests two possible approaches.

First, brokers can offer their clients the option to pay trading fees according to the maker-taker schedule (i.e., maker rebate and taker fee). Because maker-taker investors earn higher average profits than flat-fee investors, investors will switch to the maker-taker option if offered the opportunity. In turn, those who continue to pay flat fees will find wide-spread limit orders targeting only flat-fee investors less attractive because fewer investors will trade against these quotes.

Second, exchanges can reduce the maker-taker split. Investors trade off earning the larger spread on orders that target only flat-fee investors against the lower fill rate of these orders. When the maker-taker split declines, so does the difference between the two sets of the equilibrium quotes. Because the fill rate is not directly affected by the split, the wide-spread quotes become relatively less attractive.

We show that when the fraction of flat-fee investors is sufficiently low or when the maker-taker fee split is sufficiently small, flat-fee investors post the same narrow-spread quotes as the maker-taker fee traders, which all investors are willing to trade against. A sufficient reduction in the flat-fee market or the maker-taker split can always restore the maximal trading volume and welfare and reduce the profit differential between investor groups to zero.

Our results provide support for the cap on active fees in place in the U.S. and Canada⁶; more recently, SEC Chair Gary Gensler has asked SEC staff to study whether the cap should reduce proportionally for any reduction in the minimum pricing increment.⁷

⁶In 2017, Canadian Securities Regulators lowered the cap on active trading fees for non-interlisted securities. <https://www.securities-administrators.ca/news/canadian-securities-regulators-to-lower-trading-fee-cap-for-non-inter-listed-securities/>

⁷See the transcript of Gary Gensler's June 8, 2022 speech: <https://www.sec.gov/news/speech/>

Our analysis has several implications for empirical research. A standard measure of the trading costs is the cum-fee effective spread, which is the bid-ask spread on filled orders plus trading fees. Because the fees that flat-fee investors pay are generally unobservable, in practice, the cum-fee effective spread is estimated using twice the taker fee as a proxy for trading fees on a round-trip transaction. Our results suggest that doing so *overestimates* the fees paid by flat-fee investors, hindering direct comparison of the trading costs across different market participant groups. We further find, however, that this empirical proxy for the effective cum-fee spread negatively correlates with welfare (as the effective cum-fee spread widens, welfare worsens). This empirical measure can therefore serve as a proxy when analyzing welfare implications.

Our paper is closest to work by Colliard and Foucault (2012) and Foucault, Kadan, and Kandel (2013), who model liquidity demand and supply decisions of symmetrically informed traders. Colliard and Foucault (2012) show that, absent frictions, the split into maker and taker fees has no economic impact, setting the focus on the total fee charged by an exchange. Frictions, however, may impact fee neutrality. Foucault, Kadan, and Kandel (2013) show that a minimum tick size prevents limit order prices from adjusting sufficiently to compensate traders for changes in the maker-taker fee split. Recent work by Lin, Swan, and Harris (2021) examines fee neutrality in the context of market fragmentation and information acquisition, arguing that the fee split impacts price efficiency.

Empirical studies have shed some light on the impact of maker-taker fees in practice. Malinova and Park (2015) find that after the introduction of maker rebated on the Toronto Stock Exchange in 2005, quoted spreads narrowed, but that the average cum-fee spread (i.e., the quoted spread plus twice the taker fee) remained unaffected, consistent with Colliard and Foucault (2012). Moreover, Skjeltorp, Sojli, and Tham (2017) support

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the predictions of Foucault, Kadan, and Kandel (2013), using exogenous changes in maker-taker fee schedules and a technological shock for liquidity takers. Cardella, Hao, and Kalcheva (2017) examine U.S equities data from 2008-2010 and document that trading volume declines in the total fee. Di Maggio, Liu, Rizova, and Wiley (2020) provide evidence that net of fees and rebates, there is no significant difference in trading costs between maker-taker and inverted exchanges.

Several studies examine the relationship between maker-taker fee systems and the price-discreteness (tick size) friction, finding that rebate pricing may function as a workaround for sub-optimal tick sizes. Chao, Yao, and Ye (2019) predict that increased tick-size constraints result in market fragmentation, where exchanges can differ based on the maker-taker fee split, which they support empirically. Comerton-Forde, Grégoire, and Zhong (2019) show that investors may migrate to taker-maker venues as a response to sub-optimal tick sizes on other exchanges. Consistent with the predictions of Foucault, Kadan, and Kandel (2013), Cox, Van Ness, and Van Ness (2019), among others, find that the 2016 SEC Tick Pilot program led to an increase in order flow to inverted venues for equities that experienced an increase in the minimum tick size. To better understand the welfare impacts of rebate access fee pricing, Riccò, Rindi, and Seppi (2021) examine the optimal access fee pricing schedule from the perspectives of an exchange, and a social planner. The authors show that the social planner's choice of fee structure depends on valuation dispersion and trading activity.

The maker-taker fee system may impact broker routing decisions, and exacerbate agency conflicts between brokers and their clients. Cimon (2020) predicts that large differences in maker rebates across trading venues lead brokers to route limit orders based on maker rebates, rather than fill rates. O'Donoghue (2022) extends Colliard and Foucault (2012) to examine how the make-take fee split impacts broker-routing decisions when brokers charge flat commissions to their clients. For a larger make-take fee split,

O'Donoghue (2022) predicts that limit order submitters post narrower quotes, and flat commissions fall. Battalio, Corwin, and Jennings (2016) provide empirical evidence that brokers route retail orders to venues that maximize the rebates that they receive, over order execution quality. The maker-taker pricing model is also related to the payment for order flow model, in the sense that both systems aim to incentivize order flow, (see, e.g., Kandel and Marx (1999), Battalio and Holden (2001), or Parlour and Rajan (2003)). Battalio, Shkilko, and Van Ness (2016) and Anand, Hua, and McCormick (2016) empirically compare market quality under maker-taker pricing and payment for order flow.

Finally, our work complements Degryse, van Achter, and Wuyts (2020), who study the impact of post-trade clearing and settlement fees. In their model, the clearing house may set a flat fee for all trades or impose different fees, depending on whether a trade was internalized. They find that the fee schedule affects the welfare of market participants, and that the optimal schedule depends on the size of the clearing fee.

1 The Model

Investors and Security. At each period t , a risk-neutral, expected utility maximizing investor enters the market willing to trade a single unit of the risky security. Investors are buyers or sellers with equal probability, and have a private valuation for the security that is equal to $V > 0$ for buyers, and $-V$ for sellers.

Trading in a Limit Order Book. Trading for the single security conducts via limit order book. An investor may place an order to buy or sell the security using a market or a limit order. An investor that enters the market in period t may trade immediately using a market order only if there is an available limit order at the exchange on the correct side of the market (e.g., if the investor wants to buy, there must be a limit sell

order available.). If no desirable limit orders are available, an investor may attempt to trade next period by placing a limit order. A buyer places a limit order at price bid_{t+1} , and a seller places a limit order at price ask_{t+1} . We assume that investors who are indifferent between market orders and limit orders choose to submit market orders. Limit orders that are not filled by the subsequent investor are canceled. In what follows, for an order submitted in period t , we denote a market buy as MB_t , a market sell as MS_t , a limit buy as LB_t , and a limit sell as LS_t .

Exchange Fees. The limit order book is maintained by an exchange, which charges time-invariant fees for executing orders; these fees depend on the order type (market or limit). We focus on the prevalent practice of “maker-taker” fees where submitters of market orders (liquidity “takers”) pay a fee $f_T > 0$ per unit, and submitters of executed limit orders (liquidity “makers”) pay a lower fee of $f_M < f_T$ per unit (in practice, this fee is often negative and makers receive a rebate of $-f_M$); the intuition for most of our results extends to the reverse scenario.⁸ To examine the impact of maker-taker pricing, we write the taker fee as $f_T = c + f$ and the maker fee as $f_M = c - f$. The “net fee” retained by the exchange then equals $f_T + f_M = 2c$, and f describes the split of the net fee into a taker fee and a maker fee. For technical reasons, we impose a bound on the exchange fees and investor valuations.

Assumption 1 (Fee Bound) *The exchange fees and investor valuations satisfy the following bound: $3f + c < V$.*

In practice, many investors access exchanges via a third party, a broker-dealer, and they pay exchange fees as part of the broker-dealer commissions, which are typically insensitive to order types.⁹ To study the impact of maker-taker fees when some investors

⁸The “inverted” pricing where takers receive a rebate and makers pay a positive fee is often referred to by industry participants as “taker-maker pricing”, as it is utilized, for instance, by NASDAQ OMX BX.

⁹See, e.g., the Interactive Brokers fees: <https://www.interactivebrokers.com/en/pricing/commissions-home.php>

pay flat fees, we examine two types of investors, $\theta_i \in \{\theta_F, \theta_{MT}\}$. Fraction $\lambda \in [0, 1]$ of investors are of type θ_F : they access the exchange through a broker and pay a flat fee for any filled order; we refer to them as “flat-fee investors.” We assume a single competitive broker who sets the flat fee $\bar{f}(\lambda)$ equal to the per type θ_F investor average of the maker and taker fees remitted to the exchange. The remaining fraction $(1 - \lambda)$ are of type θ_{MT} : they access the exchange directly, paying maker fees on filled limit orders and taker fees on market orders; we refer to them as “maker-taker investors.”

Investor Payoffs. A buyer’s payoff in period t equals the investor’s private valuation, less the price that the investor pays, plus the fee incurred. We normalize the payoff of an unfilled order to zero. A period t investor of type θ_i has the following expected payoffs to submitting a market buy order, which trades immediately at the prevailing ask price ask_t (when available), and posting a limit buy order at price bid_{t+1} :

$$\pi_t^{MB}(ask_t, \lambda, \theta_i) = V - ask_t - fee^{MB}(\lambda, \theta_i) \quad (1)$$

$$\pi_t^{LB}(bid_{t+1}, \lambda, \theta_i) = \Pr(fill \mid bid_{t+1}, \lambda) \times (V - bid_{t+1} - fee^{LB}(\lambda, \theta_i)) \quad (2)$$

where $\Pr(fill \mid bid_{t+1}, \lambda)$ is the probability that an investor’s period t limit buy order is filled in period $t + 1$ given the order’s price, bid_{t+1} , and the fraction λ of the flat-fee investors. To avoid excessive notation, we use $fee^{MB}(\lambda, \theta_i)$ and $fee^{LB}(\lambda, \theta_i)$ to describe the fees for the market and executed limit orders, respectively, paid by an investor of type θ_i . The fill probability and the flat fee depend on λ , and therefore so do the payoffs. Payoffs to sell orders are defined analogously.

2 Equilibrium

2.1 Single Investor Type

We begin by considering two extreme cases, where all investors either i) pay maker-taker fees ($\lambda = 0$), or, ii) pay flat fees ($\lambda = 1$). The model is symmetric across buyers and sellers, and we focus on the decisions of buyers, without loss of generality.

Equilibrium Characterization. Upon arriving at the market, an investor chooses the type of order to submit (market or limit), and if they are to submit a limit order, the limit price. Following Foucault (1999) and Degryse, van Achter, and Wuyts (2020), we search for a stationary, symmetric equilibrium, where investors who submit buy limit orders post them at price $bid^*(\lambda)$ such that an incoming seller is indifferent to trading with them (using a market order) or submitting their own limit order at price $ask^*(\lambda)$. This “indifference” price setting is optimal in equilibrium. A marginally higher bid price lowers the limit order payoff upon execution, without improving the fill rate. A marginally lower bid price leaves the limit order unfilled, as the incoming seller would prefer submitting their own limit order.

All investors pay maker-taker fees. We first consider the case of $\lambda = 0$, so that all investors are of type θ_{MT} , they pay taker fees on market orders and receive maker rebates on limit orders. For investors to be indifferent between market and limit orders, $ask^*(0)$ and $bid^*(0)$ must solve:

$$V - ask^*(0) - f_T = \Pr(\text{fill} \mid bid^*(0), \lambda = 0) \times (V - bid^*(0) - f_M) \quad (3)$$

$$bid^*(0) + V - f_T = \Pr(\text{fill} \mid ask^*(0), \lambda = 0) \times (ask^*(0) + V - f_M) \quad (4)$$

When prices are set so that the next period’s investor is indifferent between a market and a limit order, a limit order’s execution probability equals the probability that the arriving investor is on the “right” side of the market, which equals $1/2$. Solving equations

(3)-(4), the equilibrium prices $ask^*(0)$ and $bid^*(0)$ are:

$$ask^*(0) = \frac{V}{3} - \frac{(2f_T - f_M)}{3} = \frac{V - c}{3} - f = -bid^*(0) \quad (5)$$

Consistent with Colliard and Foucault (2012), when investors pay maker-taker fees, the maker-taker split f impacts only the “raw” quoted bid-ask spread, defined as the difference between the ask and the bid prices, but it does not affect the effective cum-fee spread, which is economically more meaningful as it accounts for the exchange fees:

$$ask^*(0) - bid^*(0) + 2f_T = \frac{2(V + 2c)}{3}. \quad (6)$$

Substituting the equilibrium prices from (5) into (1)-(2), investors’ payoffs are also independent of the fee split f :

$$\pi^{MB}(\lambda = 0) = \frac{2}{3}(V - c) = \pi^{LB}(\lambda = 0) \quad (7)$$

All investors pay a flat fee. Suppose now that $\lambda = 1$, so that all investors are of type θ_F and pay a flat fee $\bar{f}(1)$ per transaction. The flat fee is the average (per investor trade) of the fees that the broker remits to the exchange. When all investors access the exchange via the broker, this average must equal half the net fee of $f_T + f_M = 2c$ collected by the exchange. The flat fee is then $\bar{f}(1) = c$.

The equilibrium prices $ask^*(1)$ and $bid^*(1)$ solve equations analogous to (3)-(4), except that market and executed limit orders both incur fee $\bar{f}(1)$ instead of f_T and f_M . The prices are given by:

$$ask^*(1) = \frac{V - c}{3} = -bid^*(1), \quad (8)$$

Although the “raw” quoted bid-ask spread is larger for $\lambda = 1$ than $\lambda = 0$, the “cum-fee”

bid-ask spread is the same:

$$ask^*(1) - bid^*(1) + 2\bar{f}(1) = \frac{2(V + 2c)}{3} = ask^*(0) - bid^*(0) + 2f_T \quad (9)$$

Finally, since the flat fee does not depend on the maker-taker fee split f , neither do investors' payoffs or strategies.

We summarize our results in the following proposition.

Proposition 1 (Single Investor Type) *With a single investor type, $\lambda \in \{0, 1\}$, investors' equilibrium strategies, investors' payoffs, and the cum-fee bid-ask spread are independent of the maker-taker fee split f . An increase in f of $\Delta > 0$ reduces the quoted bid-ask spread by 2Δ for $\lambda = 0$, and it does not affect the quoted spread for $\lambda = 1$.*

Proposition 1 illustrates why investors' order placement decisions and payoffs are independent of the fee redistribution between takers and makers: the increase (or decrease) in f is exactly compensated for by a reduction in the quoted spread. When comparing an environment where all investors pay maker-taker fees ($\lambda = 0$) to one where all pay flat fees ($\lambda = 1$), Proposition 1 implies the following corollary.

Corollary 1 (Maker-Taker Fees vs. Flat Fees) *The quoted spread is narrower in a maker-taker fee system ($f > 0$) than in a system that charges a flat fee per trade; the cum-fee spread is the same.*

Empirically, the cum-fee spread is estimated as the raw spread plus twice the taker fee. This empirical measure overestimates the transaction costs of the investors that pay flat fees, and we discuss this further in Section 4.2.

2.2 Two Investor Types

We now turn to the general case of a market where fraction $\lambda \in (0, 1)$ of investors are of type θ_F and pay flat fees, and the remainder are of type θ_{MT} and pay maker-taker fees. Since the flat fee is the weighted average of the taker and maker fees incurred on the trades of type θ_F investors, with $f_T > f_M$ flat-fee investors always pay less for market orders ($\bar{f}(\lambda) \leq f_T$) than maker-taker investors, and they pay more for limit orders ($\bar{f}(\lambda) \geq f_M$); the inequalities are strict when flat-fee investors use both order types. Consequently, flat-fee investors are willing to use market orders for a wider range of posted prices than maker-taker investors. When submitting a limit order, an investor therefore must choose between posting it at a relatively wide bid-ask spread, so that only flat-fee investors trade against it, or at a narrow bid-ask spread, so that both groups of investors are willing to trade against it.

Equilibrium Characterization. We continue to assume that an investor submits a market order whenever she is indifferent between a market order and a limit order, provided that there is a limit order available on the relevant side of the book. We begin by characterizing an investor's limit order pricing strategy.

Similarly to the case with a single type of investor, an investor placing the limit order chooses the price such that one group of investors are exactly indifferent between trading against this limit order and posting their own limit order according to their equilibrium strategy. An investor may post a *narrow-spread* price, which makes the next period's *maker-taker investor* indifferent; in this case, the next period's flat-fee investor strictly prefers a market order. Alternatively, she may post a *wide-spread* price, which makes the next period's *flat-fee investor* indifferent; in this case, the next period's maker-taker investor strictly prefers posting a limit order.

In what follows, we focus on investors who enter the market to buy; sell decisions are symmetric. An investor's optimal limit order posting strategy, and therefore the equi-

librium bid and ask prices, depend on the fraction λ of flat-fee investors. We use $bid_n^*(\lambda)$ and $bid_w^*(\lambda)$ to denote the bid prices at which maker-taker and flat-fee investors, respectively, are indifferent between using a market sell order and posting a sell limit order according to their respective equilibrium strategies for a given λ .

Lemma 1 (Limit Order Prices and Fill Rates) *The minimum bid price at which a flat-fee investor sells using a market order is strictly lower than that for a maker-taker investor, $bid_w^*(\lambda) < bid_n^*(\lambda)$, for any λ . Limit buy orders posted at $bid_w^*(\lambda)$ and $bid_n^*(\lambda)$ execute with probabilities $\lambda/2$ and $1/2$, respectively.*

Lemma 1 formalizes the notion of narrow- and wide-spread prices, and illustrates the trade-off between limit order prices and fill rates: wider spreads offer higher ex-post profits but reduce the probability that the order executes.

We show next that maker-taker investors have a stronger preference for higher execution probability, in the sense that whenever flat-fee investors prefer to submit limit orders at narrow spreads, so do maker-taker investors. For the investor who pays fee $f_i(\lambda) \in \{f_M, \bar{f}(\lambda)\}$ for a filled limit order, let $\delta_n^w(\lambda; f_i)$ denote the difference in expected payoffs to submitting limit buy orders at $bid_w^*(\lambda)$ and at $bid_n^*(\lambda)$ (when all other investors follow their prescribed equilibrium strategies):

$$\delta_n^w(\lambda; f_i) = \frac{\lambda}{2} (V - bid_w^*(\lambda) - f_i(\lambda)) - \frac{1}{2} (V - bid_n^*(\lambda) - f_i(\lambda)). \quad (10)$$

When the payoff difference $\delta_n^w(\lambda; f_i) > 0$, the type θ_i investor prefers to post wide-spread limit orders, which target only the flat-fee investors; for $\delta_n^w(\lambda; f_i) < 0$, the investor prefers to post orders at the narrow spread in order to trade with all investors on the right side of the market. If $\delta_n^w(\lambda; f_i) = 0$, the investor is indifferent between the two choices, and he may use a mixed strategy. We characterize an investor's choice of the limit order price for a given λ by a probability function $\eta_i(\lambda)$, for $i \in \{F, MT\}$, where

an investor of type θ_i chooses the narrow-spread price $bid_n^*(\lambda)$ with probability $\eta_i(\lambda)$, and the wide-spread price bid_w^* with probability $1 - \eta_i(\lambda)$. By examining $\delta_n^w(\lambda; f_i)$ in Equation (10) as $f_i(\lambda)$ changes, we arrive at the following lemma.

Lemma 2 (Limit Order Choice) *For any fraction $\lambda \in (0, 1)$ of flat-fee investors, the equilibrium payoff difference $\delta_n^w(\lambda; \theta_i)$ between posting a limit order at a wide-spread price and at a narrow-spread price is increasing in the fee $f_i(\lambda)$ for an executed limit order.*

Lemma 2 implies that if a flat-fee investor (weakly) prefers to post limit orders at the narrow spread to trade with all investors, then this preference is strict for a maker-taker investor (i.e., $\eta_F > 0 \Rightarrow \eta_{MT} = 1$). Conversely, if a maker-taker investor (weakly) prefers to target only the flat-fee investors by posting limit orders at the wide spreads, then a flat-fee investor strictly prefers to do the same (i.e., $\eta_{MT} < 1 \Rightarrow \eta_F = 0$). Thus, in equilibrium those who pay lower fees for limit orders (receive rebates) will provide liquidity at (weakly) more favorable prices.

Any investor that finds a limit order on the correct side of the book with a limit price that is acceptable to them, will submit a market order against it. Hence, an investor's equilibrium limit order pricing strategy fully characterizes their order placement decision. The limit order choices can be described by a pair $(\eta_F(\lambda), \eta_{MT}(\lambda))$, the probabilities that investors of type θ_F and θ_{MT} post narrow-spread orders.

Flat Fee. We now derive an expression for the flat fee paid by flat-fee investors, which we have assumed reflects the average maker and taker fees that the broker remits to the exchange on their behalf. When an investor trades with a market order, the broker pays f_T ; if (and only if) an investor's posted limit order is filled, the broker pays f_M . Recall that the fill rates for these limit orders if posted at narrow and wide spreads, respectively, are $1/2$ and $\lambda/2$, by Lemma 1. Then, denoting by $\mu_F(\lambda) = \Pr(MB^* \mid \lambda, \theta_F) + \Pr(MS^* \mid \lambda, \theta_F)$ the stationary equilibrium probability of a

market order submission for flat-fee investors, the flat fee equals:

$$\bar{f}(\lambda) = \frac{\mu_F(\lambda) \cdot f_T + (1 - \mu_F(\lambda))(\frac{1}{2}\eta_F(\lambda) + \frac{\lambda}{2}(1 - \eta_F(\lambda))) \cdot f_M}{\mu_F(\lambda) + (1 - \mu_F(\lambda))(\frac{1}{2}\eta_F(\lambda) + \frac{\lambda}{2}(1 - \eta_F(\lambda)))}. \quad (11)$$

Appendix A.2 illustrates that the flat fee can be expressed as $\bar{f}(\lambda) = c + f \times \varphi(\lambda)$, where $\varphi(\lambda) \geq 0$ for all values of λ .

Equilibrium Existence: We focus on an equilibrium where investors' order submission strategies are monotone in λ , in the sense that the probability $\eta_i(\lambda)$ that investor of type θ_i submits a limit order at the equilibrium narrow spread declines in the fraction λ of flat-fee investors. We search for thresholds $\underline{\lambda}_i$ and $\bar{\lambda}_i$ for $i \in \{F, MT\}$, such that an investor of type θ_i targets all investors by submitting narrow-spread limit orders for $\lambda \in [0, \underline{\lambda}_i]$, plays a mixed strategy over the narrow- and the wide- spread limit orders for $\lambda \in (\underline{\lambda}_i, \bar{\lambda}_i)$, and targets exclusively flat-fee investors, by submitting wide-spread limit orders, for $\lambda \in [\bar{\lambda}_i, 1]$.

Lemmas 1 and 2 imply that in any equilibrium, maker-taker investors are (weakly) more likely to post narrow spreads than flat-fee investors, and that at most one type of investors is indifferent between posting the wide and the narrow spreads for any λ . Thus, if the equilibrium thresholds exist, they must satisfy: $0 \leq \underline{\lambda}_F \leq \bar{\lambda}_F \leq \underline{\lambda}_{MT} \leq \bar{\lambda}_{MT} \leq 1$. We can now state the following existence and uniqueness theorem.

Theorem 1 (Existence and Uniqueness) *For each $\lambda \in (0, 1)$, there exist unique values $bid_w^*(\lambda) < bid_n^*(\lambda) < ask_n^*(\lambda) < ask_w^*(\lambda)$ and $\eta_F(\lambda) \leq \eta_{MT}(\lambda)$; and there exist unique values of λ , $0 < \underline{\lambda}_F < \bar{\lambda}_F < \underline{\lambda}_{MT} < \bar{\lambda}_{MT} < 1$, such that the following strategies constitute an equilibrium.*

For each $\lambda \in (0, 1)$, maker-taker (flat-fee) investors with valuation V submit buy market orders if there is a sell limit order at a price of $ask_n(\lambda)$ ($ask_w^(\lambda)$) or lower in the book. Otherwise, they submit limit buy orders as follows:*

- (Type 1) For $\lambda \in [0, \underline{\lambda}_F]$, $\eta_F(\lambda) = \eta_{MT}(\lambda) = 1$: all investors post $bid_n(\lambda)$.
- (Type 2) For $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$, $0 < \eta_F(\lambda) < \eta_{MT}(\lambda) = 1$: maker-taker investors post $bid_n(\lambda)$; flat-fee investors post $bid_n(\lambda)$ with probability $\eta_F(\lambda)$ and $bid_w(\lambda)$ with probability $1 - \eta_F(\lambda)$. Further, $\eta'_F(\cdot) < 0$.
- (Type 3) For $\lambda \in [\bar{\lambda}_F, \underline{\lambda}_{MT}]$, maker-taker investors post at $bid_n(\lambda)$, and flat-fee investors post at $bid_w(\lambda)$.
- (Type 4) For $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$, $0 = \eta_F(\lambda) < \eta_{MT}(\lambda) < 1$: maker-taker investors post $bid_n(\lambda)$ with probability $\eta_{MT}(\lambda)$ and $bid_w(\lambda)$ with probability $1 - \eta_{MT}(\lambda)$; flat-fee investors post $bid_w(\lambda)$. Further, $\eta'_{MT}(\cdot) < 0$.
- (Type 5) For $\lambda \in [\bar{\lambda}_{MT}, 1]$, all investors post $bid_w(\lambda)$.

Seller decisions of investors with valuation $-V$ are symmetric to the above.

We provide a qualitative illustration of the equilibrium types in Figure 1.

[Figure 1 about here]

3 Liquidity Provision by Maker-Taker Investors

Theorem 1 illustrates that there are five types of equilibria. In all but type 1 equilibria, i.e. for all $\lambda > \underline{\lambda}_F$, maker-taker investors are more likely than flat-fee investors to submit a limit order at the narrow spread, and they therefore enjoy a higher execution probability on their limit orders; in a type 1 equilibrium both groups post at narrow spreads. Furthermore, for $\lambda > \underline{\lambda}_F$, limit orders at prices $ask_w(\lambda)$ and $bid_w(\lambda)$ are posted with a positive probability, which implies that a maker-taker investor is less likely to submit a market order and more likely to submit a limit order, relative to a flat-fee investor. We summarize our observations in the following corollary.

Corollary 2 (Maker-taker investors as market makers) *Compared to flat-fee investors, maker-taker investors are less likely to submit a market order and more likely to submit a limit order. On average, they post limit orders at narrower spreads, and they achieve higher probability of execution on their limit orders.*

In practice, traders that pay exchange fees and rebates on a trade-by-trade basis, as maker-taker investors do in our model, have so-called direct access contracts with trading venues, and these traders often possess superior technology. Most market makers have direct access, and their superior ability to provide liquidity is typically attributed to their technological advantages. Corollary 2, however, identifies another channel: when some investors pay flat commissions through their brokers, which is common practice, traders who access exchanges directly necessarily assume a de facto market-making role, even if they do not possess a technological advantage over other investors.

3.1 Investor Profits

We examine next how the fraction λ of flat-fee investors affects the distribution of the gains from trade across the two investor groups. The difference between the average expected profits for maker-taker investors and flat-fee investors is:

$$\Delta\pi_F^{MT}(\lambda) = \mathbb{E}[\pi(\lambda \mid \theta_{MT})] - \mathbb{E}[\pi(\lambda \mid \theta_F)], \quad (12)$$

where $\mathbb{E}[\pi(\lambda \mid \theta_i)]$ is the ex ante expected payoff for an investor of type $\theta_i \in \{\theta_F, \theta_{MT}\}$, given fraction λ of flat-fee investors; this payoff is computed prior to observing the state of the limit order book or the investors' valuation.

Proposition 2 (Flat Fee: Winners and Losers) *The expected profits for maker-taker investors: i) equal those of flat-fee investors in the type 1 equilibrium, $\Delta\pi_F^{MT}(\lambda) = 0$ for*

$\lambda \in (0, \underline{\lambda}_F]$, and *ii*) exceed the profits of flat-fee investors for other equilibria types, $\Delta\pi_F^{MT}(\lambda) > 0$ for $\lambda \in (\underline{\lambda}_F, 1)$.

Proposition 2 implies that flat-fee investors (at least weakly) prefer to switch to a maker-taker fee structure, even when their broker sets the flat fee competitively. The broker's redistribution of maker-taker fees creates a disparity of profits between maker-taker and flat-fee investors, benefiting the former.

The disparity in profits between maker-taker and flat-fee investors stems from the difference in fees on limit orders: the lower maker fee ensures maker-taker investors earn greater profits than flat-fee investors for an identically-priced limit order. Hence, when facing an empty book, a maker-taker investor always earns higher expected profits than a flat-fee investor. While a flat-fee investor earns higher profit when submitting a market order against a limit order priced at ask_n , on average, this benefits is outweighed by the flat-fee investor lower profits on limit orders. In an empirical study of high-frequency market-making using recent Canadian data, Korajczyk and Murphy (2019) find that the profitability from a market-making strategy is linked to the maker-taker model, as their profits accrue most heavily (and sometimes entirely) through liquidity rebates.

3.2 Trading Volume

We measure trading volume by the probability $\mu(\lambda)$ of a trade in a given period; that is, the probability that an arriving investor submits a market order:

$$volume = \mu(\lambda) = \lambda\mu_F(\lambda) + (1 - \lambda)\mu_{MT}(\lambda), \quad (13)$$

where $\mu_i(\lambda)$, $i \in \{F, MT\}$, is the stationary equilibrium probability that an investor of type θ_i submits a market order, given the fraction λ of flat-fee investors.

A flat-fee investor submits a market order whenever the previous period's investor

was on the other side of the market and posted a limit order, and a maker-taker investor additionally requires that this limit order be posted at a narrow spread. Omitting dependence on λ to ease the exposition, in a stationary equilibrium, the market order submission probabilities are determined by the following equations:

$$\mu_F = \frac{1}{2}\lambda(1 - \mu_F) + \frac{1}{2}(1 - \lambda)(1 - \mu_{MT}) \quad (14)$$

$$\mu_{MT} = \frac{1}{2}\lambda\eta_F(1 - \mu_F) + \frac{1}{2}(1 - \lambda)\eta_{MT}(1 - \mu_{MT}) \quad (15)$$

From the proof of Theorem 1 in Appendix A.2, we show that Equation (13) via Equations (14)-(15) can be expressed as a function of η_i , $i \in \{F, MT\}$. We can then write *volume* as:

$$volume = \frac{2\lambda + \eta_F\lambda(1 - \lambda) + \eta_{MT}(1 - \lambda)(2 - \lambda)}{4 + 2\lambda - \eta_F\lambda(1 - \lambda) + \eta_{MT}(1 - \lambda)(2 + \lambda)}. \quad (16)$$

As we illustrate in Figure 2a, trading volume is non-monotonic in the fraction λ of flat-fee investors. Volume is highest when the limit order fill rate is highest, that is, when all limit orders are posted at narrow spreads (Type 1 equilibrium), as all investors are willing to trade against them. Volume is also highest in the opposite extreme, in a market in which only includes flat-fee investors (i.e, $\lambda = 1$, or a “single-tier system”).

For other equilibria types, which emerge for $\lambda \in [\underline{\lambda}_F, 1]$, however, trading volume is lower, which is a consequence of two opposing effects in λ . First, for a higher probability of a flat-fee investor arrival (higher λ), limit orders posted at wide spreads face higher execution probability. Second, as λ increases, the probability that a limit order is posted with a narrow spread declines, and consequently so does the probability that a maker-taker investor uses a market order. The second effect dominates when at least one investor type is using a mixed-strategy (Type 2 and Type 4 equilibria), while the first effect dominates when all investors employ pure strategies (Type 1, 3 and 5 equilibria).

[Figure 2 about here]

We summarize our prediction on trading volume in the proposition below.

Proposition 3 (Trading Volume) *Trading volume is maximized for $\lambda \in [0, \underline{\lambda}_F]$ (type 1 equilibrium) and at $\lambda = 1$ (in the absence of maker-taker investors).*

Our result on trading volume is consistent with Degryse, van Achter, and Wuyts (2020), who show in their analysis of clearing and settlement costs that an equilibrium where investors submit quotes that target all investors has the highest rate of trade.

3.3 Aggregate Welfare

A reduction in trading volume implies that fewer investors realize gains from trade. To analyze the impact of λ on investor welfare, in the sense of allocative efficiency, we use the average per period gain from trade, $W(\lambda)$, as in studies such as Bessembinder, Hao, and Zheng (2015). In any given period t , the investors' gain from trade depends on the types of the limit and the market order submitter, and it is not affected by the transaction price, which is simply a transfer between investors. Omitting the dependence on λ once again to ease notation, the expected gain from trade each period equals:

$$\begin{aligned} W = & \frac{1}{2}\lambda(1-\lambda)(1-\mu_F)\eta_F(2V - f_T - \bar{f}) + \frac{1}{2}\lambda(1-\lambda)(1-\mu_{MT})(2V - \bar{f} - f_M) \\ & + \frac{1}{2}\lambda^2(1-\mu_F)(2V - 2\bar{f}) + \frac{1}{2}(1-\lambda)^2(1-\mu_{MT})\eta_{MT}(2V - f_T - f_M) \end{aligned} \quad (17)$$

Collectively, the terms in Equation (17) account for all (weighed) combinations of period t and period $t - 1$ investor-type pairs that yield a trade in period- t . For instance, the first term reflects the expected gain from a trade between a limit order of a flat-fee investor (who arrived in period $t - 1$) and a market order of a maker-taker investor (who arrived in period t). This arrival occurs with probability $\lambda(1 - \lambda)$; for the trade to

take place, the flat-fee investor must post a limit order at the narrow spread (probability $(1 - \mu_F)\eta_F$) and the maker-taker investor must be on the correct side of the market to trade against it (probability $1/2$).

Because the exchange always receives $f_T + f_M = 2c$ per trade, the net fees per trade must be $-2c$. Secondly, investors that participate in a trade earn their private values, V , so the total gains from each trade is fixed at $2(V - c)$. Finally, in any period, a trade occurs with probability $\mu(\lambda)$, implying that the effects of λ on welfare are directly proportional to that of trading volume, yielding the following corollary,

Corollary 3 (Welfare) *Investor welfare is proportional to the probability of a trade, $W(\lambda) = 2\mu(\lambda)(V - c)$, and thus it co-moves with trading volume.*

Proposition 3 and Corollary 3 illustrate that welfare is highest when the execution probability of limit orders is maximized. This result is consistent with empirical findings of Hollifield, Miller, Sandås, and Slive (2006), who find that unfilled limit orders account for 70% of all welfare loss in their sample.

4 Impact of the Maker-Taker Fee Asymmetry

With a single-tier fee system, Colliard and Foucault (2012) show that the maker-taker fee split f has no re-distributive impact; thus trading volume, equilibrium profits, and investor welfare are unaffected. Proposition 1 concurs with these findings.¹⁰ In this section, we consider the implications of the maker-taker split for investor order placement strategies in a two-tier fee system. How does the redistribution impact market quality, and investor welfare? To aid in our exposition, we prove the following lemma.

¹⁰For a study of the total fee c , we refer the reader to Colliard and Foucault (2012).

Lemma 3 (Fee asymmetry f and equilibria types) *Given λ , threshold values $\{\underline{\lambda}_F, \bar{\lambda}_F, \underline{\lambda}_{MT}, \bar{\lambda}_{MT}\}$ decrease in f . In a Type 2 equilibrium, $\frac{\partial \eta_F}{\partial f} < 0$. In a Type 4 equilibrium, $\frac{\partial \eta_{MT}}{\partial f} < 0$.*

Lemma 3 highlights that the maker-taker split indirectly impacts an investor's limit order pricing strategy: for a given fraction of flat-fee investors, a reduction in f leads all investors to (weakly) widen their limit order quotes.

4.1 Trading Volume and Welfare.

In Section 2, we show that investor welfare is proportional to trading volume, and that trading volume is highest when investors target all incoming investors with their limit orders (Type 1 equilibrium). If, however, the proportion λ is fixed such that the equilibrium is *not* of Type 1, can one calibrate the maker-taker fee split to induce a Type 1 equilibrium? In the following proposition, we highlight a key prediction of our model: that f can influence the equilibrium type for a given λ .

Proposition 4 (Fee Asymmetry, Trading Volume, and Gains from Trade) *Given λ , for an increase in f , trading volume and investor welfare:*

- *declines for an equilibrium of type 2 or 4.*
- *is unaffected for an equilibrium of type 1, 3, or 5.*

Moreover, for any $\lambda \in [0, 1]$ and $f^ \in [0, \frac{V-c}{3}]$ such that $\underline{\lambda}_F(f^*) < \lambda$, there exists an $\hat{f} \leq f^*$ such that $\underline{\lambda}_F(\hat{f}) \geq \lambda$.*

The core finding of Proposition 4 is that f impacts trading volume and welfare only *indirectly* through investor strategies. Hence, any measurable impact requires a change in f that induces investors to change their limit order price strategy (i.e., change in mixed strategy probabilities). We predict that an increase in f leads an investor to

widen their quoting strategy (Lemma 3), leading trading volume to fall and welfare to worsen.

Proposition 4 also produces important guidance for policy. The model predicts that, if the equilibrium type that obtains from λ and f is not of type 1, one can reduce the level of the maker-taker fee split to induce a type 1 equilibrium, thereby improving investor welfare. A reduction in f decreases the difference in the fee schedules of maker-taker and flat-fee investors, leading flat-fee investors to prefer submitting limit orders priced at ask_n^* and bid_n^* for higher values of λ . At these limit prices, maker-taker investors do not pass over limit orders submitted by flat-fee investors, improving fill rates of limit orders placed by flat-fee investors, and subsequently volume and investor welfare. Ultimately, investors who must trade through a broker prefer to narrow the gap between the fees they pay for limit orders, and those paid by maker-taker investors.

We note here that, in the case of a monopolistic, profit-maximizing exchange, the interests of flat-fee investors and the exchange align. In our environment, the exchange pays no costs, and collects a maker fee and a taker fee for each completed trade. Thus, the maximum (average) profit exchange makes per period is the probability that a trade occurs, plus the fees they collect:

$$E[\pi_{exchange}] = \Pr(\text{trade}) \times (f_T + f_M) = \mu(\lambda) \times 2c \quad (18)$$

which is equal to a re-scaling of trading volume by $2c$. Hence, a monopolistic exchange would prefer to eliminate the maker-taker pricing scheme (set $f = 0$) in the presence of a broker who charges a flat-fee $\bar{f}$ for orders they route to the exchange.

Proposition 4 suggests that disincentivizing maker-taker investors (i.e., low f) from outbidding flat-fee investors is most beneficial from a welfare perspective. By corollary, inducing maker-taker investors to become de facto market-makers (i.e., type 5 equilibrium) through a larger f has the opposite effect. By performing a purely market-making

role, maker-taker investors now pass over *any* limit orders in the book to provide their own, eliminating all trades initiated by maker-taker investors. While flat-fee investors benefit from the increased instances of liquidity provision by maker-taker investors, the former effect dominates. In aggregate, increasing the maker-taker fee split impacts *flat-fee investors* positively by improving liquidity provision from maker-investors, but this positive impact is outweighed by the negative impact on limit order fill rates for *all investors*.

As a policy tool, a regulator who wishes to maximize welfare could induce a Type 1 equilibrium by influencing an exchange's choice of f by imposing, for example, an appropriate cap on the re-distributive split, f . To see this, we show in the proof of Theorem 1 that $\underline{\lambda}_F$ is given by,

$$\underline{\lambda}_F = \frac{4(V - c)/3 - f}{4(V - c)/3 + f/2} \quad (19)$$

Then, as $f \rightarrow 0$, $\underline{\lambda}_F \rightarrow 1$, which implies that there must exist an $\hat{f}$ such that $\lambda < \underline{\lambda}_F(\hat{f})$ for any $\lambda \in (0, 1)$. Hence, a regulator can always impose a cap on f to induce a Type 1 Equilibrium for any investor group split (λ) in a two-tiered system. While a cap of $f = 0$ trivially accomplishes the task, a positive cap is possible, provided that the regulator takes into account the market share of traders who do not access the exchange directly.

4.2 Quoted, Effective, and Cum-Fee Spreads.

Exchanges argue that maker rebates incentivize investors to increase the provision of cheaper liquidity, citing tighter quoted bid-ask spreads as a positive result of the maker-taker fee system.¹¹ The raw quoted spread, $ask - bid$, measures the average price of limit orders posted by investors. In our model, symmetry simplifies this measure to twice the

¹¹For specific examples, see TMX Group (04-05-2015) and BATS Global Markets Inc. (06-01-2015); for a recent general discussion on maker-taker fees and market quality, see the Securities and Exchange Commission Division of Trading and Markets (20-10-2015) Memorandum.

average price posted by an investor. Here, we look at the “unconditional investor” (that is, not conditional on type). We calculate the average quoted spread by computing a weighting of all quotes by their expected submission rate. We denote the quoted spread as $spread_j$, where j indicates the equilibrium type (i.e., $i \in \{1, 2, \dots, 5\}$).

$$spread_j = 2 \times \left(\frac{\Pr(\text{LO at } ask_n)}{\Pr(\text{LO})} \times ask_n + \frac{\Pr(\text{LO at } ask_w)}{\Pr(\text{LO})} \times ask_w \right) \quad (20)$$

Different from the quoted spread, we also consider the effective spread, which measures the spread of the average quotes at which trades actually occur. Symmetry simplifies this measure to twice the average price paid by an investor, independent of their type. We calculate the average price by computing a weighting of all quotes by their expected trading rate. We denote the effective spread as $espread_j$, where j indicates the equilibrium type (i.e., $i \in \{1, 2, \dots, 5\}$).

$$espread_j = 2 \times \left(\frac{\Pr(\text{MO against } ask_n)}{\mu(\lambda)} \times ask_n + \frac{\Pr(\text{MO against } ask_w)}{\mu(\lambda)} \times ask_w \right) \quad (21)$$

While the effects of f on the average quoted and effective spreads are overall non-monotonic, the impact on both is monotonic within the equilibrium regions. Moreover, we find that the average quoted spread is always wider than the average effective spread.

Proposition 5 (Quoted and Effective Spreads) *Given λ , an increase in f ,*

- *narrows the quoted and effective spread within any pure strategy equilibrium.*
- *widens the quoted and effective spread within any mixed-strategy equilibrium.*

Moreover, the quoted spread is wider than the effective spread for all f .

[Figure 3 about here]

Proposition 5 underscores an important effect of the maker-taker fee split when investors face different fee schedules: when flat fee investors quote at a wider price on average

than maker-taker investors, their limit orders are passed over more often, which leads the effective spread (i.e., the spread paid by takers) to be narrower, on average, than the observed quoted spread (i.e., the spread posted by makers). We illustrate this point in Figure 3a. Proposition 5 also suggests that any marginal changes to f *within* any pure strategy equilibrium are asynchronous to the impacts on volume and welfare (the quoted and effective spreads narrow, while volume and welfare are unchanged).

Though the effective spread does not account for all trading costs borne by investors, we find that the impact of f on the effective cum-fee spread—a measure of the spread that includes trading fees—behaves similarly across most pure-strategy equilibria (Fig. 3b, dash line). We write the round-trip effective cum-fee spread as $espread_j + 2 \times f_i$, where f_i is the take fee paid by the investor (f_T for maker-taker investors, $\bar{f}$ for flat-fee investors), and j is the equilibrium type. We measure the effective cum-fee spread as a volume-weighted average, similar to our effective spread measure. Denote the average effective cum-fee spread as ACF_j .

$$ACF_j = espread_j + 2 \times \left(\frac{\mu_F \lambda}{\mu(\lambda)} \times \bar{f} + \frac{\mu_{MT}(1 - \lambda)}{\mu(\lambda)} \times f_T \right) \quad (22)$$

In the proposition below, we document how an increase in the maker-taker split impacts ACF_j , for each equilibrium type, j .

Proposition 6 (Average Effective Cum-Fee Spread) *Given λ , an increase in f :*

- *narrows the average effective cum-fee spread for equilibrium types 1 and 3.*
- *widens the average effective cum-fee spread for equilibrium types 2, 4 and 5.*

Unlike the raw effective spread, we find that the impact on the average effective cum-fee spread *within* pure-strategy equilibria is not universally narrowing. When investors target all investors with their limit orders (Type 1), the average effective cum-fee spread declines because the raw effective spread narrows by $2f$, but fees increase by less than

$2f$, because flat-fee investors pay less than the taker fee for an executed market order. Conversely, when all investors target flat-fee investors only (Type 5), an increase in f raises the flat-fee investor's fee by $2 \times \varphi(\lambda)$, while the raw effective spread narrows by $2 \times \varphi(\lambda) \cdot \frac{2-\lambda}{2+\lambda}$. Comparatively, the impact across the regions of pure strategy equilibria show that as f increases, investors switch their limit order pricing strategy to quote less favorable prices (and as a result, target only flat-fee investors). The result is an increase in the average effective cum-fee spread.

Empirically, the above (true) effective cum-fee spread is difficult to measure, as the fees paid by flat-fee investors are unobservable. Instead, the effective cum-fee spread is commonly estimated by adding the cost of twice the taker fee to the effective spread, $spread_j + 2 \times f$. This practice tends to overestimate the cost to flat-fee investors. However, our model suggests that measuring the effective cum-fee spread in this way aligns qualitatively with our predictions on trading volume and welfare. We denote the estimated average effective cum-fee spread as $ECF_j = spread_j + 2 \times f$, where i describes the type of equilibrium.

Proposition 7 (Estimated Effective Cum-Fee Spreads) *Given λ , ECF widens in f within any equilibrium type, and $ECF_1 \leq ECF_2 \leq ECF_3 \leq ECF_4 \leq ECF_5$. Moreover, $ECF_j \geq ACF_j \geq spread_j$ for all $j \in \{1, \dots, 5\}$, and is negatively correlated with volume and welfare.*

That the estimated effective cum-fee spread increases in f (see Figure 3b, dash-dot line) provides an additional testable prediction that co-moves with our result on welfare. Together, our model predicts that as the effective cum-fee spread widens in f , trading volume should fall, which we posit has a negative effect on investor welfare.

5 Conclusion

In limit order markets, exchanges often set access fee schedules that discriminate between liquidity makers and liquidity takers. While we concur with Colliard and Foucault (2012) that the split between the maker and taker fee is irrelevant when all investors face the same fee schedule, we argue that this irrelevance breaks down in a two-tiered fee system. In a model where some investors pay maker-taker fees directly and others pay flat fees through a broker, the split of the total exchange fee into the maker fee and the taker fee plays a meaningful role—even when the maker-taker fees are passed through, on average, to those investors paying a flat-fee.

In a two-tiered system, flat-fee investors price their limit orders (weakly) less favorably than maker-taker investors, in response to the higher fees they face when trading with limit orders. As such, maker-taker investors pass over their limit orders to submit their own, leading to a reduction trading volume; this effect amplifies with the maker-taker fee differential. Moreover, we show that investor welfare co-moves with trading volume, thereby worsening in a two-tiered system. The welfare loss is disproportionately borne by flat-fee investors, who earn relatively lower profits on average, compared to maker-taker investors. Together, our results provide support for the industry practice of capping the redistributive split, usually by a cap on the taker fee. We also predict that the common empirical measure of the effective cum-fee spread negatively co-moves with volume, yielding a key testable prediction that a decrease in the fee split leads to a reduction in observable effective cum-fee spreads, and thus, an improvement in volume and welfare.

Finally, we contribute to the debate on incentives to provide liquidity in limit order markets. Our model suggests that providing maker-taker pricing to a subset of investors may incentivize them to take on a more prominent market-making role—a result that is

independent of the relative speed of investors. While these de facto market makers may use their rebate advantage to quote more favorably, their enhanced competitiveness in liquidity provision simultaneously dissuades them from initiating trades, *and* worsens the competitiveness of flat-fee investors as liquidity providers. As such, our paper cautions that incentives for liquidity provision offered to one subset of investors may not pass down liquidity improvements to all.

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A Appendix

A.1 List of Variables and Parameters

Variable	Description
V	private valuation of trade to an investor.
c	per-leg transaction cost collected by the exchange.
f_T	fee charged by the exchange for filled market orders.
f_M	fee charged by the exchange for filled limit orders.
f_{total}	sum of f_T and f_M .
$\bar{f}$	(average) flat fee charged by broker.
λ	total mass of investors who pay a flat fee per trade.
θ_i	denotes an investor's type, $i \in \{\theta_{MT}, \theta_F\}$.
t	subscript to denote a period t value.
ask_w	ask price posted at "wide spread".
ask_n	ask price posted at "narrow spread".
MB_t (MS_t)	period t investor submits a market buy (sell) order.
LB_t (LS_t)	period t investor submits a limit buy (sell) order.
π_i^j	profit of order type $j \in \{MB, MS, LB, LS\}$ to a type i investor.
$\delta_n^w(\lambda; f_i)$	measures $\pi_i^{LB}(bid_w^*) - \pi_i^{LB}(bid_n^*)$ for a type i investor that pays f_i .
$\underline{\lambda}_F$	λ value below which type θ_F investors submit only a narrow spread.
$\bar{\lambda}_F$	λ value above which type θ_F investors submit only a wide spread.
$\underline{\lambda}_{MT}$	λ value below which type θ_{MT} investors submit only a narrow spread.
$\bar{\lambda}_{MT}$	λ value below which type θ_{MT} investors submit only a wide spread.
$\varphi(\lambda)$	average redistribution value f paid by the broker to the exchange.
$\Delta\Pi$	denotes $\pi_{MT}(\lambda) - \pi_F(\lambda)$.
$\mu(\lambda)$	the probability of a trade.
μ_F	the probability of a market order submitted by a type θ_F investor.
μ_{MT}	the probability of a market order submitted by a type θ_{MT} investor.
W	aggregate investor welfare.
$spread_j$	quoted spread for equilibrium region j .
$espread_j$	effective spread for equilibrium region j .
ACF_j	average effective cum-fee spread for equilibrium region j .
ECF_j	empirical effective cum-fee spread for equilibrium region j .

A.2 Proofs in Section 2

Preliminaries. In the proofs that follow, we simplify $\bar{f}$ from (11) to:

$$\bar{f} = \frac{\mu_F(\lambda) \cdot f_T + (1 - \mu_F(\lambda))(\frac{\lambda}{2}(1 - \eta_F(\lambda)) + \frac{1}{2}\eta_F(\lambda)) \cdot f_M}{\mu_F(\lambda) + (1 - \mu_F(\lambda))(\frac{\lambda}{2}(1 - \eta_F(\lambda)) + \frac{1}{2}\eta_F(\lambda))} \quad (23)$$

$$= \frac{\mu_F(\lambda) - (1 - \mu_F(\lambda))(\frac{\lambda}{2}(1 - \eta_F(\lambda)) + \frac{1}{2}\eta_F(\lambda))}{\mu_F(\lambda) + (1 - \mu_F(\lambda))(\frac{\lambda}{2}(1 - \eta_F(\lambda)) + \frac{1}{2}\eta_F(\lambda))} \times f + c = \varphi(\lambda) \times f + c, \quad (24)$$

where $\varphi(\lambda)$ represents the average f paid by the broker to the exchange. We can solve $\varphi(\lambda)$ as a function of η_{MT} , η_F and λ explicitly by inserting the solutions to μ_{MT} and μ_F from (14)-(15):

$$\varphi(\lambda) = \frac{(2 - \lambda - \lambda^2) - (1 + \lambda - 2\lambda^2)\eta_F - (1 - 2\lambda + \lambda^2)\eta_{MT} \times \eta_F}{(2 + \lambda + \lambda^2) + (1 - \lambda)\eta_F + 2(\lambda - \lambda^2) \cdot \eta_{MT} + (1 - 2\lambda + \lambda^2)\eta_{MT} \times \eta_F}. \quad (25)$$

Proof (Lemma 1). Consider a buyer ($V > 0$). A buyer that pays flat fees is indifferent to trading against a limit order priced at ask_w^* , and submitting a limit order priced at bid_n^* . Because $ask_n^* < ask_w^*$, trading against a limit order priced at ask_n^* is preferred to a limit order priced at ask_w^* , and thus a flat fee investor would trade against any limit orders priced at ask_n^* or ask_w^* . Buyers who pay to maker-taker fees are indifferent to trading against a limit order priced at ask_n^* , and submitting a limit order priced at bid_n^* . However, because $ask_n^* < ask_w^*$, a maker-taker investor prefers submitting a limit order at price bid_n^* over trading against a limit order priced at ask_w^* . Hence, a maker-taker investor only trades against limit orders priced at ask_n^* . Thus, a (weakly) larger measure of buyers will trade against limit orders priced at ask_n^* than will trade against limit orders priced at ask_w^* . The argument is similar for investors who sell. ■

Proof (Lemma 2). We begin by taking the difference, $\Delta(\lambda, f) = \delta_n^w(\lambda; \bar{f}) - \delta_n^w(\lambda; f_M)$, and simplify to yield:

$$\Delta = \delta_n^w(\lambda, \bar{f}) - \delta_n^w(\lambda, f_M) = \frac{\lambda}{2}(-\psi f - f) - \frac{1}{2}(-\psi f - f) = \frac{(1 - \lambda)}{2}(1 + \phi)f. \quad (26)$$

Note that $\psi < 1$ by the fact that $\bar{f} > f_M \iff (\psi f + c) > (c - f)$. ■

Proof (Theorem 1). We begin by deriving the flat fee, and we then focus on a single equilibrium type in each step. As a last step, we verify that the ranges of λ do not overlap and that each value of λ corresponds to a single equilibrium type. We characterize the buyers' decisions, seller decisions are symmetric.

Step 1: Flat fee. We begin by deriving the probabilities for market and limit order submissions, which in turn define the flat fee that investors pay. Omitting dependence on λ to simplify notation, when maker-taker and flat-fee investors submit limit orders at narrow spreads with probabilities (η_{MT}, η_F) , respectively, their probabilities of market order submission, $\mu_{MT} = 2\Pr(MB_{MT}) = 2\Pr(MS_{MT})$ and $\mu_F = 2\Pr(MB_F) = 2\Pr(MS_{MT})$ are determined by the following system:

$$\mu_{MT} = \frac{1}{2}\lambda\eta_F(1 - \mu_F) + \frac{1}{2}(1 - \lambda)\eta_{MT}(1 - \mu_{MT}) \quad (27)$$

$$\mu_F = \frac{1}{2}\lambda(1 - \mu_F) + \frac{1}{2}(1 - \lambda)(1 - \mu_{MT}), \quad (28)$$

where we use the fact that for a maker-taker investor to submit a market order in equilibrium, there must be a limit order at a narrow spread on the correct side of the book, whereas a flat-fee investor submits a market order as long as the previous investor posted a limit order on the correct side, at either equilibrium price. Solving these:

$$\mu_{MT} = \frac{\eta_F(\lambda + \lambda^2) + \eta_{MT}(2 - \lambda - \lambda^2)}{4 + 2\lambda - \eta_F(\lambda - \lambda^2) + \eta_{MT}(2 - \lambda - \lambda^2)} \quad (29)$$

$$\mu_F = \frac{2 - \eta_F(\lambda - \lambda^2) + \eta_{MT}(\lambda - \lambda^2)}{4 + 2\lambda - \eta_F(\lambda - \lambda^2) + \eta_{MT}(2 - \lambda - \lambda^2)}, \quad (30)$$

For each set of the order submission probabilities (η_{MT}, η_F) , the flat fee can be computed from (11), using the above (29)-(30).

Step 2: Type 1 equilibrium. When all investors submit narrow-spread prices,

the narrow-spread bid and ask price, which make the maker-taker investor indifferent between a market and a limit order, must satisfy:

$$V - ask_n^*(\lambda) - f_T = \frac{1}{2}(V - bid_n^*(\lambda) - f_M). \quad (31)$$

By symmetry, $ask_n^*(\lambda) = -bid_n^*(\lambda)$, and we obtain:

$$ask_n^*(\lambda) = \frac{1}{3}(V - c) - f = -bid_n^*(\lambda). \quad (32)$$

To obtain the wide-spread price, we first observe that the probabilities of market and limit order submissions and therefore the flat fee in a Type 1 equilibrium are the same as in the environment with a single investor type, because in equilibrium all investors submit market orders as long as there is a limit order on the correct side of the book. The probabilities of market order submission (29)-(30) simplify to $\mu_{MT} = \mu_F = 1/3$, and therefore $\bar{f}(\lambda) = c$. The wide-spread ask price satisfies the indifference condition for the flat-fee investor, whose equilibrium limit order strategy prescribes posting at the narrow spreads:

$$V - ask_w^*(\lambda) - \bar{f}(\lambda) = \frac{1}{2}(V - bid_n^*(\lambda) - \bar{f}(\lambda)). \quad (33)$$

Substituting the values for $bid_n^*(\lambda)$ and $\bar{f}(\lambda)$, we obtain:

$$ask_w^*(\lambda) = ask_n^*(\lambda) + \frac{3}{2}f = \frac{1}{3}(V - c) + \frac{1}{2}f = -bid_w^*(\lambda). \quad (34)$$

Finally, for a flat-fee investor to prefer posting at the narrow-spread prices, fraction λ of the flat-fee investors must satisfy:

$$\delta_n^w(\lambda, \bar{f}(\lambda)) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - \bar{f}(\lambda)) - \frac{1}{2}(V - bid_n^*(\lambda) - \bar{f}(\lambda)) \leq 0. \quad (35)$$

Using expressions (32) and (34) for the bid prices and $\bar{f}(\lambda) = c$, inequality (35) can be rewritten as:

$$g_F(\lambda) \equiv \frac{4\lambda - 1}{3\lambda + 2}(V - c) + \frac{1}{2}f \leq 0. \quad (36)$$

The unique threshold $\underline{\lambda}_F$, which solves (36) with equality so that a flat-fee investor is indifferent between posting the wide and the narrow spread prices (32) and (34), is then defined and bounded as follows:

$$\underline{\lambda}_F = \frac{4(V - c)/3 - f}{4(V - c)/3 + f/2} > \frac{(V - c)}{4(V - c)/3 + 1/6(V - c)} = \frac{2}{3}, \quad (37)$$

where the inequality follows by Assumption 1.

By Lemma 2, a maker-taker investor earns higher profit by posting at the narrow spread; further, one can verify that a maker-taker investor that arrives prefers posting a limit order to trading with a market order at the wide spread.

Step 3: Type 3 equilibrium. When maker-taker-investors post narrow spreads and flat-fee investors post wide spreads, the indifference condition for a maker-taker investor is the same as in the Type 1 equilibrium, given by (31), and the narrow-spread bid and ask prices are given by (32). The indifference condition for a flat-fee investor becomes:

$$V - ask_w^*(\lambda) - \bar{f}(\lambda) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - \bar{f}(\lambda)). \quad (38)$$

Using symmetry of the buy and sell decision, the wide-spread prices are:

$$ask_w^*(\lambda) = \frac{2 - \lambda}{2 + \lambda}(V - \bar{f}(\lambda)) = -bid_w^*(\lambda). \quad (39)$$

Using expressions (11) and (29)-(30), together with $(\eta_{MT}, \eta_F) = (1, 0)$, the flat fee in the

Type 3 equilibrium is given by:

$$\bar{f}(\lambda) = c + f \times \frac{(1 - \lambda)(2 + \lambda)}{2 + 3\lambda - \lambda^2}. \quad (40)$$

For the two groups of investors to follow the prescribed limit order submission strategies, we must have, for the bid prices in (32) and (39) and flat fee in (40):

$$\delta_n^w(\lambda, \bar{f}(\lambda)) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - \bar{f}(\lambda)) - \frac{1}{2}(V - bid_n^*(\lambda) - \bar{f}(\lambda)) \geq 0 \quad (41)$$

$$\delta_n^w(\lambda, f_M) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - f_M) - \frac{1}{2}(V - bid_n^*(\lambda) - f_M) \leq 0. \quad (42)$$

Thresholds $\bar{\lambda}_F$ and $\underline{\lambda}_{MT}$ are the values of λ that satisfy, respectively, (41) and ((42) with equality. Substituting (32) and (39) in and rearranging, thresholds $\bar{\lambda}_F$ and $\underline{\lambda}_{MT}$ are the values of λ that solve conditions (43) and (44), respectively:

$$\bar{g}_F(\lambda) \equiv \frac{4\lambda - 1}{3} \frac{\lambda - 1}{2 + \lambda}(V - c) + \frac{2 - \lambda + \lambda^2}{2 + 3\lambda - \lambda^2} f \geq 0 \quad (43)$$

$$\underline{g}_{MT}(\lambda) \equiv \frac{4\lambda - 1}{3} \frac{\lambda - 1}{2 + \lambda}(V - c) + \frac{\lambda^2(3 - \lambda)}{2 + 3\lambda - \lambda^2} f \leq 0. \quad (44)$$

We next show that there exist unique thresholds $\bar{\lambda}_F$ and $\underline{\lambda}_{MT}$ that satisfy (43) and ((44) with equality and that also satisfy $\underline{\lambda}_F < \bar{\lambda}_F < \underline{\lambda}_{MT} < 1$.

Existence. First, observe that at $\lambda = 1$, $\bar{g}_F(1) > 0$ and $\underline{g}_{MT}(1) > 0$. Second, at $\lambda = \underline{\lambda}_F$:

$$\bar{g}_F(\underline{\lambda}_F) = \bar{g}_F(\underline{\lambda}_F) - \underline{g}_F(\underline{\lambda}_F) = \frac{3}{2} \frac{(2/3 - \underline{\lambda}_F)(1 - \underline{\lambda}_F)}{2 + 3\underline{\lambda}_F - \underline{\lambda}_F^2} f < 0, \quad (45)$$

where we used $\underline{g}_F(\underline{\lambda}_F) = 0$ and the inequality follows because $2/3 < \underline{\lambda}_F < 1$, by (37).

Consequently, there exists $\bar{\lambda}_F \in (\underline{\lambda}_F, 1)$ that solves $\bar{g}_F(\bar{\lambda}_F) = 0$. Third, at $\lambda = \bar{\lambda}_F$:

$$\underline{g}_{MT}(\bar{\lambda}_F) = \underline{g}_{MT}(\bar{\lambda}_F) - \bar{g}_F(\bar{\lambda}_F) = \frac{(1 - \bar{\lambda}_F)(\bar{\lambda}_F - 2)(\lambda + 1)}{2 + 3\bar{\lambda}_F - \bar{\lambda}_F^2} f < 0. \quad (46)$$

Consequently, there exists $\underline{\lambda}_{MT} \in (\bar{\lambda}_F, 1)$ that solves $\underline{g}_{MT}(\underline{\lambda}_{MT}) = 0$.

Uniqueness. Threshold $\underline{\lambda}_{MT}$ is unique because function $\underline{g}_{MT}(\bar{\lambda}_F)$ is strictly increasing in λ on $(0,1)$; this can be shown by direct differentiation of the coefficients on $(V - c)$ and f , which are both increasing. Further, $\underline{g}'_{MT}(\cdot) > 0$ also ensures the correct sign for the inequality (42) for $\lambda \in (\bar{\lambda}_F, \underline{\lambda}_{MT})$. To prove uniqueness for $\bar{\lambda}_F$, observe that for $\lambda \in (0, 1)$ inequality (43) is equivalent to

$$\bar{g}_F(\lambda)(2 + \lambda)(2 + 3\lambda - \lambda^2) \geq 0. \quad (47)$$

The left-hand side of (47) is a cubic polynomial with a negative first coefficient ($= -4(V - c)/3 + f$), which can have at most three roots and can cross 0 from below at most once. Our existence argument demonstrates that at $\lambda = \bar{\lambda}_F \in (\underline{\lambda}_F, 1)$, this polynomial must cross 0 from below. Therefore the solution on $(\underline{\lambda}_F, 1)$ is unique, and furthermore, the inequality has the desired sign on $(\bar{\lambda}_F, \underline{\lambda}_{MT})$.

Step 4: Type 5 equilibrium. When all investors post wide spreads, the indifference condition for a flat-fee investor is the same as in the Type 3 equilibrium, given by (38), and the wide-spread bid and ask prices are given by (39). The flat fee in this setting is derived using (11) and (29)-(30), together with $(\eta_{MT}, \eta_F) = (0, 0)$:

$$\bar{f}(\lambda) = c + f \times \frac{(1 - \lambda)(2 + \lambda)}{2 + \lambda + \lambda^2}. \quad (48)$$

Solving the following indifference condition for a maker-taker investor

$$V - ask_n^*(\lambda) - f_M = \frac{\lambda}{2}(V - bid_w^*(\lambda) - \bar{f}(\lambda)), \quad (49)$$

the narrow-spread equilibrium price in this setting is given by:

$$ask_n^*(\lambda) = (1 - \frac{\lambda}{2})(V - c) - (1 + \frac{\lambda}{2})f - \frac{\lambda}{2}ask_w^*(\lambda). \quad (50)$$

For a maker-taker investor to submit wide-spread limit orders, we must have, for the bid prices in (39) and (50) and the flat fee $\bar{f}(\lambda)$ in (48):

$$\delta_n^w(\lambda, f_M) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - f_M) - \frac{1}{2}(V - bid_n^*(\lambda) - f_M) \geq 0. \quad (51)$$

Rearranging, we obtain the following condition on λ :

$$\bar{g}_{MT}(\lambda) \equiv \frac{4}{3} \frac{\lambda - 1}{2 + \lambda}(V - c) + \frac{2\lambda^2}{2 + \lambda + \lambda^2}f \geq 0. \quad (52)$$

By Lemma 2, if maker-taker investors post wide spreads, flat-fee investors do the same. Therefore, for the existence of the Type 5 equilibrium, it suffices to show that there exists threshold $\bar{\lambda}_{MT}$ such that condition (52) is satisfied. To provide existence, similarly to the previous step, we first observe that for $\lambda = 1$, $\bar{g}_{MT}(1) > 0$. Next, at $\lambda = \underline{\lambda}_{MT}$:

$$\bar{g}_{MT}(\underline{\lambda}_{MT}) = \bar{g}_{MT}(\underline{\lambda}_{MT}) - \underline{g}_{MT}(\underline{\lambda}_{MT}) = \frac{\underline{\lambda}_{MT}^2(\underline{\lambda}_{MT} - 1)^2(\underline{\lambda}_{MT} - 2)}{2(2 + \underline{\lambda}_{MT} + \underline{\lambda}_{MT}^2)(2 + 3\underline{\lambda}_{MT} - \underline{\lambda}_{MT}^2)} < 0. \quad (53)$$

Consequently, there exists $\bar{\lambda}_{MT} \in (\underline{\lambda}_{MT}, 1)$ that satisfies (52) with equality. Further, this value is unique because $\bar{g}_{MT}(\cdot)$ is strictly increasing; it can be shown by direct differentiation that the coefficients on $(V - c)$ and f are both increasing in λ . The positive derivative also ensures the correct sign in (52) for $\lambda \in (\bar{\lambda}_{MT}, 1)$.

Step 5: Type 2 equilibrium. We now turn to the mixed strategy equilibria. By Lemma 2, when a flat-fee investor is willing to mix (and thus indifferent between posting the wide-spread and the narrow-spread prices), a maker-taker investor strictly prefers to post at the narrow spread; and when a maker-taker investor mix, a flat-fee investor strictly prefer to post at the wide spread. To show the existence of the equilibrium, it therefore suffices to consider only the investor that mixes.

When a flat-fee investor to mixes, as prescribed by the Type 2 equilibrium, for $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$, the maker-taker investor indifference condition is the same as in the Type 1 and Type 3 equilibria, given by (31), and the narrow-spread bid and ask prices are given by (32). For a flat-fee investor to mix, we must have:

$$V - ask_w^*(\lambda) - \bar{f}(\lambda) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - \bar{f}(\lambda)) = \frac{1}{2}(V - bid_n^*(\lambda) - \bar{f}(\lambda)), \quad (54)$$

where the first equality defines the wide-spread price, and the second equality signifies the flat-fee investor indifference between the two limit order strategies. Consequently, the wide-spread price is given by the same expression as in the Type 3 equilibrium (equation (39)). To determine the flat fee function for this range of λ , we first compute the probabilities of the market order submissions (29)-(30), for $\eta_{MT}(\lambda) = 1$:

$$\mu_{MT}(\lambda) = \frac{2 - (1 - \eta_F(\lambda))(\lambda + \lambda^2)}{6 + (1 - \eta_F(\lambda))(\lambda - \lambda^2)} \quad (55)$$

$$\mu_F(\lambda) = \frac{2 + (1 - \eta_F(\lambda))(\lambda - \lambda^2)}{6 + (1 - \eta_F(\lambda))(\lambda - \lambda^2)}, \quad (56)$$

Substituting these into (11) and omitting dependence of η_F on λ to ease notation:

$$\bar{f}(\lambda) = c + f \times \varphi(\lambda) = c + f \times \frac{(1 - \eta_F)(2 + \lambda)(1 - \lambda)}{2(1 + \eta_F) + (1 - \eta_F)(3\lambda - \lambda^2)}. \quad (57)$$

Now, by substituting prices (32) and (39) and the flat fee (57) into the indifference con-

dition (54), the probability of submitting the narrow-spread order for a flat-fee investor must satisfy:

$$\frac{4\lambda - 1}{3} \frac{\lambda - 1}{2 + \lambda} (V - c) + \frac{2 + (1 - \eta_F)\lambda^2 - (1 - \eta_F)\lambda}{2(1 + \eta_F) + (1 - \eta_F)(3\lambda - \lambda^2)} f = 0. \quad (58)$$

Solving equation (59) yields unique $\eta_F \in (0, 1)$ that solves it for each $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$:

$$\begin{aligned} \eta_F &= \frac{2 + 3\lambda - \lambda^2}{\lambda - 1} \frac{\bar{g}_F(\lambda)}{\underline{g}_F(\lambda)(2 - \lambda) + (3\lambda/2 - 1)f} \\ &= 1 - \frac{4}{1 - \lambda} \frac{\underline{g}_F(\lambda)}{\underline{g}_F(\lambda)(2 - \lambda) + (3\lambda/2 - 1)f}, \end{aligned} \quad (59)$$

where $\underline{g}_F(\lambda)$ and $\bar{g}_F(\lambda)$ are as defined in Steps 2 and 3 of the proof. The denominator in (59) is positive on $(\underline{\lambda}_F, \bar{\lambda}_F)$ because $\underline{g}_F(\lambda) > 0$ for $\lambda > \underline{\lambda}_F$, and $\lambda > \underline{\lambda}_F > 2/3$. Expressions (59) illustrate that $\eta_F = 1$ at $\lambda = \underline{\lambda}_F$ and $\eta_F = 0$ at $\lambda = \bar{\lambda}_F$. To show that η_F decreases in λ on $(\underline{\lambda}_F, \bar{\lambda}_F)$, we apply the Implicit Function theorem, using equation (58). We write equation (58) as follows:

$$\underline{h}(\lambda, \eta_F) \equiv g(\lambda)dn_F(\lambda, \eta_F) + nm_F(\lambda, \eta_F) = 0, \quad (60)$$

where $g(\lambda) = \underline{g}_F(\lambda)/f - 1/2$, $dn_F(\lambda, \eta_F)$ and $nm_F(\lambda, \eta_F)$ denote the denominator and the numerator of the second term in (58), respectively. Taking partial derivatives:

$$\frac{\partial \underline{h}}{\partial \lambda} = g'(\lambda)dn_F(\lambda, \eta_F) + g(\lambda)(1 - \eta_F)(3 - 2\lambda) + 2(1 - \eta_F)(2\lambda - 1) > 0 \quad (61)$$

$$\frac{\partial \underline{h}}{\partial \eta_F} = g(\lambda)(2 - 3\lambda + \lambda^2) - \lambda^2 + \lambda = \frac{\underline{g}_F(\lambda)}{f}(2 - \lambda)(1 - \lambda) + (1 - \lambda)\left(\frac{3\lambda}{2} - 1\right) > 0. \quad (62)$$

We use the following properties for $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$ to sign the inequalities: $g(\lambda) < 0$,

$g'(\lambda) > 0$, $\underline{g}_F(\lambda) > 0$, $dn_F(\lambda, \eta_F) > 0$, and $\underline{\lambda}_F > 2/3$. By the Implicit Function theorem, $\eta'_F(\lambda) = -(\partial \underline{h}/\partial \lambda)/(\partial \underline{h}/\partial \eta_F) < 0$.

Step 6: Type 4 equilibrium. Similarly to the Type 2 equilibrium, we only need consider only the investor that mixes. When a maker-taker investor to mixes, as prescribed by the Type 4 equilibrium, for $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$, the flat-fee investor posts wide spreads, and his indifference condition is the same as in the Type 3 equilibria, and the wide-spread bid and ask prices are given by (39) (but the flat fee differs). For a maker-taker investor to mix, we must have:

$$V - ask_n^*(\lambda) - f_M = \frac{1}{2}(V - bid_n^*(\lambda) - f_M) = \frac{\lambda}{2}(V - bid_w^*(\lambda) - f_M). \quad (63)$$

The first equality implies that the narrow-spread bid and ask prices are given by (32), as in the Types 1-3 equilibria. To determine the flat fee function for this range of λ , we first compute the probabilities of the market order submissions (29)-(30), for $\eta_F(\lambda) = 0$:

$$\mu_{MT}(\lambda) = \frac{(1 - \lambda)\eta_{MT}(\lambda)}{2 + (1 - \lambda)\eta_{MT}(\lambda)} \quad (64)$$

$$\mu_F(\lambda) = \frac{1}{\lambda + 2} \times \frac{2 + \lambda(1 - \lambda)\eta_{MT}(\lambda)}{2 + (1 - \lambda)\eta_{MT}(\lambda)}. \quad (65)$$

Substituting these into (11) and omitting dependence of η_{MT} on λ to ease notation:

$$\bar{f}(\lambda) = c + f \times \frac{(2 + \lambda)(1 - \lambda)}{2 + \lambda + \lambda^2 + 2\eta_{MT}(\lambda - \lambda^2)}. \quad (66)$$

Substituting prices (32) and (39) and the flat fee (66) into the indifference condition (63), the probability of submitting the narrow-spread order for a maker-taker investor must satisfy:

$$\frac{4\lambda - 1}{3} \frac{\lambda - 1}{2 + \lambda} (V - c) + \frac{\lambda^2(2 + (1 - \lambda)\eta_{MT})}{2 + \lambda + \lambda^2 + 2\eta_{MT}(\lambda - \lambda^2)} f = 0. \quad (67)$$

Solving equation (67) yields unique $\eta_F \in (0, 1)$ that solves it for each $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$:

$$\begin{aligned}\eta_{MT} &= \frac{\lambda^2 + \lambda + 1}{2\lambda(\lambda - 1)} \times \frac{\bar{g}_{MT}(\lambda)}{\underline{g}_{MT}(\lambda) + 2\lambda f/(2 + 3\lambda - \lambda^2)} \\ &= 1 - \frac{2 + 3\lambda - \lambda^2}{2\lambda(1 - \lambda)} \times \frac{\underline{g}_{MT}(\lambda)}{\underline{g}_{MT}(\lambda) + 2\lambda f/(2 + 3\lambda - \lambda^2)},\end{aligned}\quad (68)$$

where $\underline{g}_{MT}(\lambda)$ and $\bar{g}_{MT}(\lambda)$ are as defined in Steps 3 and 4 of the proof. The denominator in (68) is positive on $(\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$ because $\underline{g}_{MT}(\lambda) > 0$ for $\lambda > \underline{\lambda}_{MT}$. Expressions (68) illustrate that $\eta_{MT} = 1$ at $\lambda = \underline{\lambda}_{MT}$ and $\eta_{MT} = 0$ at $\lambda = \bar{\lambda}_{MT}$. To show that η_{MT} decreases in λ on $(\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$, we apply the Implicit Function theorem, using equation (67). We write equation (67) as follows:

$$\bar{h}(\lambda, \eta_{MT}) \equiv g(\lambda)dn_{MT}(\lambda, \eta_{MT}) + nm_{MT}(\lambda, \eta_{MT}) = 0, \quad (69)$$

where $g(\lambda) = \underline{g}_F(\lambda)/f - 1/2$, $dn_{MT}(\lambda, \eta_{MT})$ and $nm_{MT}(\lambda, \eta_{MT})$ denote the denominator and the numerator of the second term in (67), respectively. Taking partial derivatives:

$$\frac{\partial \bar{h}}{\partial \lambda} = g'(\lambda)dn_{MT}(\lambda, \eta_{MT}) + g(\lambda)(1 + 2\eta_{MT} + 2\lambda - 4\eta_{MT}\lambda) + \lambda(4 + (2 - 3\lambda)\eta_{MT}) \quad (70)$$

$$\frac{\partial \bar{h}}{\partial \eta_{MT}} = 2\lambda(1 - \lambda)(g(\lambda) + \lambda) = 2\lambda(1 - \lambda)(\underline{g}_{MT}(\lambda)/f + 2\lambda/(2 + 3\lambda - \lambda^2)), \quad (71)$$

Both partial derivatives are positive. To sign them, we use the following properties for $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$: $g(\lambda) < 0$, $g'(\lambda) > 0$, $\underline{g}_{MT}(\lambda) > 0$, $dn_{MT}(\lambda, \eta_F) > 0$. By the Implicit Function theorem, $\eta'_{MT}(\lambda) = -(\partial \bar{h}/\partial \lambda)/(\partial \bar{h}/\partial \eta_{MT}) < 0$. ■

A.3 Proofs in Section 3

Proof (Proposition 3). We begin by reproducing (16) below:

$$\mu(\lambda) = \frac{2\lambda + \eta_{MT} \cdot (2 - 3\lambda + \lambda^2) + \eta_F \cdot (\lambda - \lambda^2)}{2(2 + \lambda) + \eta_{MT} \cdot (2 - \lambda - \lambda^2) - \eta_F \cdot (\lambda - \lambda^2)}. \quad (72)$$

By setting $\lambda = 0$ (and hence, $\eta_{MT} = \eta_F = 1$), we see that the left limit of (72) is equal to $1/3$. Similarly, setting $\lambda = 1$ generates the same result. The proposition then follows by showing that Equation (72) is less than $1/3$ for all $\lambda \in (\underline{\lambda}_F, 1)$, and equal to $1/3$ on $\lambda \in (0, \underline{\lambda}_F]$. Solving this inequality for zero, we obtain:

$$\mu(\lambda) \leq \frac{1}{3} \iff -4(1 - \lambda) \times (1 - \eta_{MT}(1 - \lambda) - \eta_F \lambda) \leq 0. \quad (73)$$

Now, let $\lambda \in [0, \underline{\lambda}_F]$. Then, we have $(\eta_{MT}, \eta_F) = (1, 1)$. The left-hand side of the inequality in equation (73) is then equal to zero, showing that on $[0, \underline{\lambda}_F]$, volume is equal to $1/3$. Next, note that for $\eta_F \in [0, 1)$ and $\lambda \in (\underline{\lambda}_F, 1)$, it must be that $\eta_{MT}(1 - \lambda) + \eta_F \lambda < 1$, implying that the inequality in (73) is strict. Thus, trading volume is always lower when $\lambda \in (\underline{\lambda}_F, 1)$ than on $\lambda \in [0, \underline{\lambda}_F] \cup \{1\}$. ■

Proof (Proposition 2). We prove this proposition in six steps: We compare the profits of flat-fee investors to maker-taker investors for the five equilibrium type, and step 6 shows that the profits of flat-fee investors are highest in the type 1 equilibrium. The buy and sell payoffs are symmetric, and we focus on investors that want to buy. We omit dependence on λ whenever possible without ambiguity.

Step 1: type 1 equilibrium. Consider $\lambda \in [0, \underline{\lambda}_F]$. All investors post narrow quotes ask_n^* and bid_n^* and $\eta_F(\lambda = \eta_{MT}(\lambda) = 1$. In this equilibrium, $\pi^{MB}(ask_n^*, \theta_{MT}) = \pi^{LB}(bid_n^*, \theta_{MT})$, and we write the difference in expected profits $\Delta\pi_F^{MT}$ defined in (12) as

$$\Delta\pi_F^{MT} = \pi^{MB}(ask_n^*, \theta_{MT}) - \mu_F \pi^{MB}(ask_n^*, \theta_F) - (1 - \mu_F) \pi^{LB}(bid_n^*, \theta_F). \quad (74)$$

By the proof of Theorem 1, the probability of submitting a market order is the same for both investor types, $\mu_F = \mu_{MT} = 1/3$, the flat fee equals c , and the probability that a posted limit order is filled is $1/2$. Using the equilibrium prices (31) in (74), we obtain:

$$\begin{aligned} \Delta\pi_F^{MT} &= V - ask_n^* - c - f - \frac{1}{3} \times (V - ask_n^* - c) - \frac{2}{3} \cdot \frac{1}{2} \times (V + ask_n^* - c) \\ &= \frac{1}{3} (V - c) - f - ask_n^* = 0. \end{aligned} \quad (75)$$

Thus, for all $\lambda \in [0, \underline{\lambda}_F]$, $\Delta\pi_F^{MT}(\lambda) = 0$.

Step 2: type 2 equilibrium. Consider $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$. Maker-taker investors post only narrow quotes, $\eta_{MT}(\lambda) = 1$, and flat-fee investors who submit limit orders post narrow quotes with probability $\eta_F(\lambda) \in (0, 1)$. In this equilibrium, a flat-fee investor earns the same payoff for all outcomes, except for trading with a market order at a narrow spread: $\pi^{LB}(bid_n^*, \theta_F) = \pi^{LB}(bid_w^*, \theta_F) = \pi^{MB}(ask_w^*, \theta_F)$. Furthermore, because maker-taker investors only use market orders when the spreads are narrow, the probability that a flat-fee investor trades with a market order at a narrow spread equals $\mu_{MT}(\lambda)$. Finally, limit orders at narrow spreads execute with probability $1/2$.

Using these insights, we can rewrite the difference in expected profits $\Delta\pi_F^{MT}$ in terms of the narrow-spread bid and ask prices ask_n^* and bid_n^* , and we further simplify it to:

$$\Delta\pi_F^{MT} = \mu_{MT}(\bar{f} - f_T) + (1 - \mu_{MT}) \frac{1}{2} (\bar{f} - f_M) \quad (76)$$

$$= \frac{1}{2} ((1 - 3\mu_{MT}) + (1 + \mu_{MT})\varphi) \times f > 0, \quad (77)$$

where $\varphi(\lambda) > 0$ is defined by expression (57) for the flat fee $\bar{f}(\lambda)$, and $\mu_{MT}(\lambda) < 1/3$ by Corollary 2. Thus, for all $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$, $\Delta\pi_F^{MT}(\lambda) > 0$.

Step 3: type 3 equilibrium. Consider $\lambda \in [\bar{\lambda}_F, \underline{\lambda}_{MT}]$. Maker-taker investors post only narrow quotes, $\eta_{MT}(\lambda) = 1$, and flat-fee investors post only wide quotes, $\eta_F(\lambda) = 0$. By construction, flat-fee investors earn the same payoffs to market and limit orders at wide spreads: $\pi^{LB}(bid_w^*, \theta_F) = \pi^{MB}(ask_w^*, \theta_F)$. Similarly to Step 2, the probability that a flat-fee investor trades with a market order at a narrow spread equals $\mu_{MT}(\lambda)$. Finally, limit orders at narrow spreads execute with probability $1/2$, and limit orders at wide spread execute with probability $\lambda/2$. The difference in expected profits $\Delta\pi_F^{MT}$ equals:

$$\Delta\pi_F^{MT} = \mu_{MT}(\bar{f} - f_T) + (1 - \mu_{MT}) \left(\frac{1}{2}(V - bid_n^* - f_M) - \frac{\lambda}{2}(V - bid_w^* - \bar{f}) \right) \quad (78)$$

To show that $\Delta\pi_F^{MT} > 0$, we rearrange (78) using $f_M = c - f$, $f_T = c + f$, the flat fee $\bar{f}(\lambda)$ from (40), the narrow and wide spread equilibrium prices from (32) and (39), and $\mu_{MT}(\lambda)$ from (15), which in a type 3 equilibrium simplifies to $\mu_{MT}(\lambda) = (1 - \lambda)/(3 - \lambda)$. After some algebraic simplification, we can rewrite (78) as:

$$\Delta\pi_F^{MT} = \frac{4}{2} \frac{1 - \lambda}{2 + \lambda} (V - c) > 0. \quad (79)$$

Thus, for all $\lambda \in [\bar{\lambda}_F, \underline{\lambda}_{MT}]$, $\Delta\pi_F^{MT}(\lambda) > 0$.

Step 4: type 4 equilibrium. Consider $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$. Maker-taker investors who submit limit orders post narrow quotes with probability $\eta_{MT}(\lambda) \in (0, 1)$, and flat-fee investors post only wide quotes, $\eta_F(\lambda) = 0$. In this equilibrium, a maker-taker investor earns the same payoff for all equilibrium actions: $\pi^{LB}(bid_n^*, \theta_{MT}) = \pi^{LB}(bid_w^*, \theta_{MT}) = \pi^{MB}(ask_n^*, \theta_{MT})$. Similarly to the previous steps, $\pi^{LB}(bid_w^*, \theta_F) = \pi^{MB}(ask_w^*, \theta_F)$, and the probability that a flat-fee investor trades with a market order at a narrow spread equals $\mu_{MT}(\lambda)$. Limit at wide spread execute with probability $\lambda/2$.

Using these insights, we can rewrite the difference in expected profits $\Delta\pi_F^{MT}$ in terms of market order payoffs to trading an narrow spreads and limit order payoffs to trading

at wide spreads, and we further simplify it to:

$$\Delta\pi_F^{MT} = \mu_{MT}(\bar{f} - f_T) + (1 - \mu_{MT})\frac{\lambda}{2}(\bar{f} - f_M). \quad (80)$$

Using explicit expressions for f_T , f_M , the probability $\mu_{MT}(\lambda)$ from Equation (15), and the flat-fee $\bar{f}(\lambda)$ from (66), the profit difference is proportional to:

$$\begin{aligned} \Delta\pi_F^{MT} &\propto -\eta_{MT}(1 - \lambda)(\lambda + \lambda^2 + \eta_{MT}(\lambda - \lambda^2)) + \lambda(2 + \eta_{MT}(\lambda - \lambda^2)) \\ &\propto 2 - \eta_{MT}(1 - \lambda)(1 + \eta_{MT}(1 - \lambda)) > 0. \end{aligned} \quad (81)$$

The inequality obtains because $\eta_{MT}(\lambda)(1 - \lambda) \in (0, 1)$ for $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$ and the values of $2 - y(1 + y)$ are > 0 for all $y \in (0, 1)$.

Step 5: type 5 equilibrium. Consider $\lambda \in [\bar{\lambda}_{MT}, 1]$. Here, flat-fee investors only trade with market orders that trade against quotes at ask_w^* , or with limit orders placed at bid_w^* . Maker-taker investors only submit limit orders priced at bid_w^* , and never use market orders. The profit difference $\Delta\pi_F^{MT} > 0$ by Lemma 2, which implies that $\pi^{LB}(bid_w^*, \theta_{MT}) > \pi^{LB}(bid_w^*, \theta_F)$, for all λ .

Step 6: flat-fee investors' preference for type 1 equilibrium. In this step, we show that flat-fee investors' profits are highest in a type 1 equilibrium, for any level of the total exchange fee c .

First, a maker-taker investor earns the same expected profit of $\pi^{MB}(ask_n^*, \theta_{MT}) = 2/3(V - c)$ for all $\lambda \in [0, \bar{\lambda}_{MT})$ (equilibria types 1-4): this investor is indifferent between trading with a market order at the narrow spread and any other outcome, and the narrow spread prices for these equilibria types do not depend on λ .

Second, a flat-fee investor earns the same profit as a maker-taker investor in a type-1 equilibrium, and earns a lower profit than a maker-taker investor, and thus lower than their own type-1 equilibrium profit, in equilibria of types 2-4.

What remains to be shown is that a flat-fee investor's profit is below $2/3(V - c)$ in the type-5 equilibrium. We compute this profit directly, using expressions for the wide-spread prices (39) and (48) for the flat fee and rearranging:

$$\pi^{MB}(ask_w^*, \theta_F) = \pi^{LB}(bid_w^*, \theta_F) = \frac{2\lambda}{2 + \lambda}(V - c) - f \times \frac{2\lambda(1 - \lambda)}{2 + \lambda + \lambda^2} \leq \frac{2}{3}(V - c), \quad (82)$$

where the inequality is strict for all $\lambda < 1$. To obtain this result, note that the equality holds for $\lambda = 1$, and that the right-hand side of (82) increases in λ for $\lambda \in (\bar{\lambda}_{MT}, 1)$ (the λ -term on $(V - c)$ increases in λ ; the λ -term on $(-f)$ decreases in λ for $\lambda > \sqrt{2} - 1$, and $\bar{\lambda}_{MT} > 2/3 > \sqrt{2} - 1$). ■

A.4 Proofs in Section 4

Proof (Lemma 3). We first show that the equilibrium thresholds $\{\underline{\lambda}_F, \bar{\lambda}_F, \underline{\lambda}_{MT}, \bar{\lambda}_{MT}\}$ decline in f , and we then show the result for the probabilities (η_F, η_{MT}) of the narrow-spread limit orders.

Equilibrium thresholds and f . Expression (37) provides a closed-form expression for the threshold $\underline{\lambda}_F$; taking a partial derivative $\partial \underline{\lambda}_F / \partial f \propto -2(V - c) < 0$.

To show that $\bar{\lambda}_F$ is decreasing in f , we recall equation (43).

$$\bar{\lambda}_F : \frac{4}{3} \frac{\bar{\lambda}_F - 1}{2 + \bar{\lambda}_F} (V - c) + \frac{2 - \bar{\lambda}_F + \bar{\lambda}_F^2}{2 + 3\bar{\lambda}_F - \bar{\lambda}_F^2} f = 0 \quad (83)$$

As in the proof of Theorem 1 The solution λ to this equation admits $\bar{\lambda}_F$. Employing the notation simplification in (83) that $f = \frac{(V-c)x}{3}$, where $x \in (0, 1)$, we use the implicit

function theorem on Equation (83) to arrive at:

$$\bar{\lambda}'_F(f) = -\frac{3}{2(V-c)} \frac{(2 - \bar{\lambda}_F + \bar{\lambda}_F^2)(2 + 3\bar{\lambda}_F - \bar{\lambda}_F^2)(2 + \bar{\lambda}_F)^2}{\left(16(1-x) + 8 + 72\bar{\lambda}_F + (30 + 16x)\bar{\lambda}_F^2 - 4(9-2x)\bar{\lambda}_F^3 + (6+x)\bar{\lambda}_F^4\right)} \quad (84)$$

We show that Equation (84) is negative by first noting that the denominator is positive by inspection. In the numerator, we can show this term is positive by noting that the only negative term is $-4(9-2x)\bar{\lambda}_F^3$. Then, because $72\bar{\lambda}_F > -36\bar{\lambda}_F^3$ for all $\bar{\lambda}_F \in [\underline{\lambda}_F, 1]$, it must be that Equation (84) is always negative. Hence, $\bar{\lambda}_F$ is decreasing in f .

Finally, thresholds $\underline{\lambda}_{MT}$ and $\bar{\lambda}_{MT}$ are the unique values of λ that solve, respectively, (44) and (52) with equality. From Steps 3 and 4 of the proof of Theorem 1, the characterizing functions for $\underline{\lambda}_{MT}$ and $\bar{\lambda}_{MT}$, from (44) and (52), are strictly increasing in λ and f for $\lambda \in (0, 1)$. Consequently, thresholds $\underline{\lambda}_{MT}$ and $\bar{\lambda}_{MT}$ decrease in f , by the Implicit Function theorem.

Mixed-strategies and f . To show that the probabilities of narrow spread submission decline in the fee split f within each mixed equilibrium type, for a given lambda, we use explicit expressions (59) and (68) for η_F and η_{MT} from the proof of Theorem 1. Taking a partial derivative of η_F with respect to f , for $\lambda \in (\underline{\lambda}_F, \bar{\lambda}_F)$, and rearranging:

$$\frac{\partial \eta_F}{\partial f} \propto \left(\frac{3}{2}\lambda - 1\right) \left(\underline{g}_F - \frac{\partial \underline{g}_F}{\partial f} \times f\right) = -\frac{4}{3} \left(\frac{3}{2}\lambda - 1\right) \frac{1-\lambda}{\lambda+2} (V-c) < 0, \quad (85)$$

where $\underline{g}_F(\lambda)$ is defined by (36). The inequality obtains because $\lambda > \underline{\lambda}_F > 2/3$.

Similarly, taking a partial derivative of η_{MT} with respect to f , for $\lambda \in (\underline{\lambda}_{MT}, \bar{\lambda}_{MT})$, and rearranging:

$$\frac{\partial \eta_{MT}}{\partial f} \propto -\frac{2+3\lambda-\lambda^2}{2\lambda(1-\lambda)} \left(\frac{4}{3} \frac{1-\lambda}{\lambda+2} (V-c) + \frac{(1-\lambda)(2+\lambda)}{2+3\lambda-\lambda^2} \times f\right) < 0. \quad (86)$$

Thus, investor mixed strategies η_F and η_{MT} are decreasing in f . ■

Proof (Proposition 4). Expression (16) illustrates that the fee split f only affects volume through the probabilities η_F and η_{MT} of narrow-spread limit orders for each investor type.

To prove this proposition, we break the proof into two parts: one in which the change in f keeps the equilibrium within the same region, and one in which the change in f leads to a shift in equilibrium. What we mean here is that, as f changes, Lemma 3 implies that the thresholds in λ change. If the change in f leads to a significant enough change in the thresholds such that the equilibrium region changes (e.g., at the edges of the equilibrium regions), then the impact on volume may be more complicated.

First, we begin with the changes in f at λ such that the region does not change. For equilibrium types 1, 3 and 5, all investors play pure strategies. Therefore, if λ and changes in f are such that the equilibrium remains within the region, then volume is unaffected. For equilibrium types 2 and 4, flat-fee and maker-taker investors, respectively, play a mixed strategy. Lemma 3 yields that these mixed strategies decrease in f . For a type 2 equilibrium we substitute $\eta_{MT}(\lambda) = 1$ into (16) and differentiate with respect to f :

$$\frac{dvol}{df} \propto \lambda(1 - \lambda) \times \frac{\partial \eta_F}{\partial f} < 0, \quad (87)$$

Now, consider a type 4 equilibrium. Here, the flat-fee investors employ a pure strategy, and maker-taker investors play a mixed strategy. Hence, we substitute $\eta_F(\lambda) = 0$ into (16) and differentiate with respect to f :

$$\frac{dvol}{df} \propto (1 - \lambda)^2(2 + \lambda) \times \frac{\partial \eta_{MT}}{\partial f} < 0, \quad (88)$$

Next, because volume is proportional to welfare, the welfare results follow directly from

the preceding volume arguments.

Finally, we show that for any pair (λ, f^*) such that $\underline{\lambda}_F(f^*) < \lambda$, there exists a unique $\hat{f} < f^*$ such that $\underline{\lambda}_F(\hat{f}) = \lambda$. Recall the closed-form solution for $\underline{\lambda}_F(f)$.

$$\underline{\lambda}_F = \frac{4(V-c)/3 - f}{4(V-c)/3 + f/2} \quad (89)$$

We can see that $\underline{\lambda}_F$ is continuous, and from Lemma 3, decreasing in f . By inspection, $f = 0 \Rightarrow \underline{\lambda}_F = 1$, and $f = \frac{V-c}{3} \Rightarrow \underline{\lambda}_F = 2/3$. Then, for any $\lambda > \underline{\lambda}_F(f^*)$ (such that the initial equilibrium is not type 1), one can lower f to any $f \in [0, \hat{f}]$ to obtain a type 1 equilibrium, which by Proposition 3 implies that volume is maximized. ■

Proof (Proposition 5). To prove this proposition, we first make the substitution of $f = \frac{V-c}{3} \cdot x$ into (20) and (21). Next, we compute the first derivative of the raw quoted spread $spread_j$ within each equilibrium region, and show that this derivative is negative for the pure strategy regions $(\{1, 3, 5\})$, and positive for the mixed strategy regions $(\{2, 4\})$. For $spread_j(x)$, we have:

$$\frac{\partial spread_j(x, \lambda)}{\partial x} = \begin{cases} -\frac{V-c}{3} & \text{if } (\eta_F, \eta_{MT}) = (1, 1) \\ \frac{(V-c)q_0(\lambda, x)}{3(1-\lambda)(2+\lambda+\lambda^2)(2+\lambda)^2(2-\lambda(2+x))^2} & \text{if } (\eta_F, \eta_{MT}) = (\eta_F, 1) \\ -\frac{(4+4\lambda-7\lambda^3-\lambda^4)(V-c)}{3(2-\lambda)(1+\lambda)(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 1) \\ \frac{(V-c)q_1(\lambda, x)}{3(2+\lambda+\lambda^2)(2(6-(4x+5)\lambda)-(1+3x)\lambda^2)-(4+3x)\lambda^3+\lambda^4(2+x)(1+\lambda))^2} & \text{if } (\eta_F, \eta_{MT}) = (0, \eta_{MT}) \\ -\frac{(1-\lambda)(2-\lambda)(V-c)}{3(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 0) \end{cases} \quad (90)$$

Equation (90) is positive for equilibrium type 2 for all $\lambda > \underline{\lambda}_F = \frac{2(4-x)}{8+x}$. To show this, we first note that the denominator is positive by inspection. Then, to show that $q_0(\lambda, x)$ of the numerator is positive, we plot it across the range of $x \in [0, 1]$ and $\lambda \in (\underline{\lambda}_F(x), 1]$ in the online appendix. By graphical representation, Equation (90) is positive for all $\eta_F \in [0, 1]$ when $\eta_{MT} = 1$. In similar fashion, we detail function $q_1(\lambda, x)$ and prove that it is positive in the online appendix.

Next, we take the first derivative of the effective spread $espread_j$ within each equilibrium region, and show that this derivative is negative for the pure strategy regions $j \in \{1, 3, 5\}$, and positive for the mixed strategy regions $j \in \{2, 4\}$.

$$\frac{\partial espread_j(x, \lambda)}{\partial x} = \begin{cases} -\frac{V-c}{3} & \text{if } (\eta_F, \eta_{MT}) = (1, 1) \\ \frac{2(V-c)r_0(\lambda, x)}{3(1-\lambda)^2(2+\lambda+\lambda^2)(3\lambda-2)} & \text{if } (\eta_F, \eta_{MT}) = (\eta_F, 1) \\ -\frac{(4+3\lambda^2-8\lambda^3+\lambda^4)(V-c)}{3(2+\lambda+\lambda^2)(2-\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 1) \\ \frac{(V-c)r_1(\lambda, x)}{3(2+\lambda+\lambda^2)(8-2(2+7x)\lambda^2+2(2x+1)\lambda^3+5x\lambda^4+(6+7x)\lambda^5+(2+3x)\lambda^6)^2} & \text{if } (\eta_F, \eta_{MT}) = (0, \eta_{MT}) \\ -\frac{(1-\lambda)(2-\lambda)(V-c)}{3(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 0) \end{cases} \quad (91)$$

Equation (91) is positive for equilibrium type 2 for all $\lambda > \underline{\lambda}_F = \frac{2(4-x)}{8+x}$. To show this, we first note that the denominator is positive by inspection, as $3\lambda - 2 > 0$ for $\lambda > \underline{\lambda}_F = \frac{2(4-x)}{8+x}$. Then, to show that $r_0(\lambda, x)$ of the numerator is positive, we plot it across the range of $x \in [0, 1]$ and $\lambda \in (\underline{\lambda}_F(x), 1]$ in the online appendix. By graphical representation, Equation (91) is positive for all $\eta_F \in [0, 1]$ when $\eta_{MT} = 1$. In similar fashion, we detail function $r_1(\lambda, x)$ and prove that it is positive in the online appendix.

Finally, to show that $spread_j \geq espread_j$ for any $j \in \{1, \dots, 5\}$, we compare the two spreads for each j . We begin with the pure strategy equilibria. For $j = 1$ or $j = 5$, the result follows trivially, as investors only submit and trade at one price in both equilibria. Let $j \in \{2, 3, 4\}$. In these equilibria, both prices are possible, in equilibrium. However, for any equilibrium in which the price ask_w is part of the limit order submission strategy for some investor, these limit orders are, in equilibrium, passed over by arriving maker-taker investors with probability 1. Thus, $spread_j$, which weights the prices of *submitted* limit orders, will always have a higher weight on ask_w than $espread_j$, which only accounts for orders that are filled. Hence, $spread_j \geq espread_j$. ■

Proof (Proposition 6). To show that ACF_j is increasing in f , we differentiate ACF_j

by f for each $j \in \{1, 2, \dots, 5\}$. We have:

$$\frac{\partial ACF(x, \lambda)}{\partial x} = \begin{cases} -\frac{2\lambda^2(1+\lambda)(V-c)}{3(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (1, 1) \\ \frac{2(V-c)(4-\lambda(10-5\lambda-\lambda^3)+x\lambda(2+\lambda+2\lambda^2+\lambda^3))}{3(1-\lambda)^2(2+\lambda+\lambda^2)(3\lambda-2)} & \text{if } (\eta_F, \eta_{MT}) = (\eta_F, 1) \\ -\frac{2\lambda^2(1-\lambda)(V-c)}{3(2-\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 1) \\ \frac{2(V-c)r_2(\lambda, x)\lambda^2}{3(2+\lambda)(2+\lambda+\lambda^2)((4-x\lambda^2)(2-3\lambda)+(4+x\lambda)\lambda^3)^2} & \text{if } (\eta_F, \eta_{MT}) = (0, \eta_{MT}) \\ \frac{2\lambda(1-\lambda)(V-c)}{3(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 0) \end{cases} \quad (92)$$

It is immediate that (92) negative for cases where $(\eta_F, \eta_{MT}) = \{(1, 1), (0, 1)\}$, and positive for $(\eta_F, \eta_{MT}) = (0, 0)$. We detail function $r_2(\lambda, x)$ and prove that it is positive in the online appendix.

Equation (92) is positive for equilibrium type 2 for all $\lambda > \underline{\lambda}_F = \frac{2(4-x)}{8+x}$. To show this, we first note that the denominator is positive by inspection, as $3\lambda - 2 > 0$ for $\lambda > \underline{\lambda}_F = \frac{2(4-x)}{8+x}$. Then, to show that the third term of the numerator is positive, we plot it across the range of $x \in [0, 1]$, and $\lambda \in (\underline{\lambda}_F(x), 1]$. By graphical representation, Equation (92) is positive for all $\eta_F \in [0, 1]$ when $\eta_{MT} = 1$. ■

Proof (Proposition 7). We define $ECF_j = \text{espread}_j + 2 \times f$, where $j \in \{1, 2, \dots, 5\}$ indexes the equilibrium regions, and substitute $f = \frac{(V-c)}{3}x$, as in previous proofs. We first note that f must be increasing in x , so for the regions of espread_j that are increasing in x from the proof of Proposition 5, namely $j \in \{2, 4\}$, it follows that ECF is increasing in f within those regions. To show that the empirical effective cum-fee spread $ECF(f, \lambda)$ is increasing in f for $j \in \{1, 3, 5\}$, we differentiate ECF with respect to x , we have:

$$\frac{\partial ECF(x, \lambda)}{\partial x} = \begin{cases} 0 & \text{if } (\eta_F, \eta_{MT}) = (1, 1) \\ \frac{8(V-c)\lambda^3}{3(2-\lambda+\lambda^2)(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 1) \\ \frac{4\lambda(V-c)}{3(2+\lambda+\lambda^2)} & \text{if } (\eta_F, \eta_{MT}) = (0, 0) \end{cases} \quad (93)$$

Thus, all cases in (93) are positive, and ECF is increasing in f , for all $f \in [0, \frac{V-c}{3}]$.

Next, we show that because ECF is continuous in f , we need only show that ECF

converges in the limit around each equilibrium threshold of λ in f .

Finally, $ECF_j \geq ACF_j \geq espread_j$ by the following argument. By definition, $ACF_j \geq espread_j$, as $ACF_j = espread_j + 2(\text{average fee})$, where 'average fee' is defined as in ACF ,

$$\text{average fee} = \left(\frac{\mu_F \lambda}{\mu(\lambda)} \times \bar{f} + \frac{\mu_{MT}(1 - \lambda)}{\mu(\lambda)} \times f_T \right) \quad (94)$$

Thus, 'average fee' is positive by definition. Then, $2f_T > 2(\text{average fee})$, as $\bar{f} < f_T$.

This implies that $ECF_j = espread_j + 2f_T \geq espread_j + 2(\text{average fee}) = ACF_j$. ■

Figure 1: Equilibrium Thresholds

Panel (a) provides an illustrative sketch of the shape of functions η_F (solid line) and η_{MT} (dashed line) in λ . The dotted lines indicate equilibrium threshold values in λ . Panel (b) plots the functions for λ_F , $\bar{\lambda}_{MT}$ (solid lines), $\bar{\lambda}_F$, λ_{MT} (dashed lines) in $\lambda - x$ space, where x is such that $f = \frac{(V-c)x}{3}$. We have also marked the three pure strategy regions with their equilibrium type. Parameter values are set to $V = 1$ and $c = 0$; other values produce qualitatively similar results.

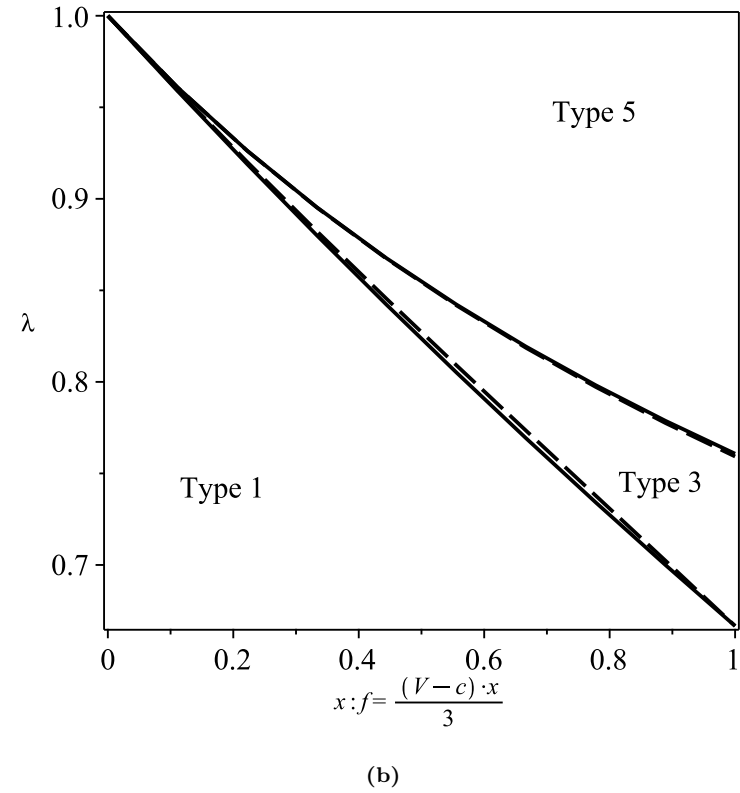
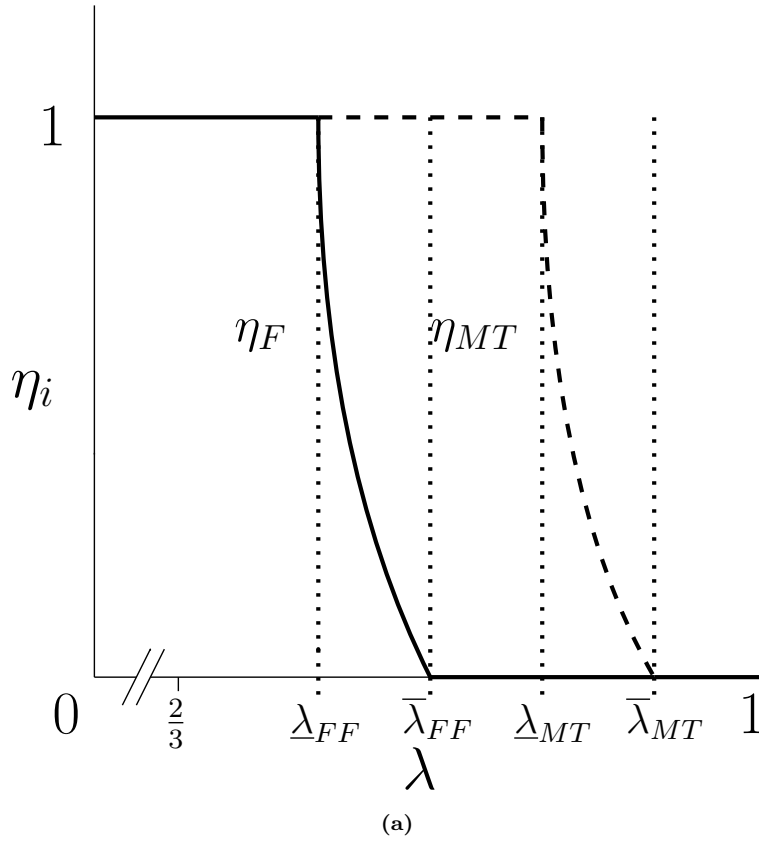


Figure 2: Trading Volume and Profit Differential

Panel (a) plots trading volume in λ ; panel (b) plots the profit functions for π_{FF} (solid line) and π_{MT} (dashed line) in λ . The dash-dot line indicates the level for the single-investor type benchmark (i.e., $\lambda \in \{0, 1\}$). Dotted vertical marked 'w' through 'z' indicate the threshold values $\underline{\lambda}_F, \bar{\lambda}_F, \underline{\lambda}_{MT}$ and $\bar{\lambda}_{MT}$, in order. Parameter values are set to $V = 1$ and $c = 0$, and $f = 1/6$, which implies a possible total gains from each trade of 2. Other values produce qualitatively similar results.

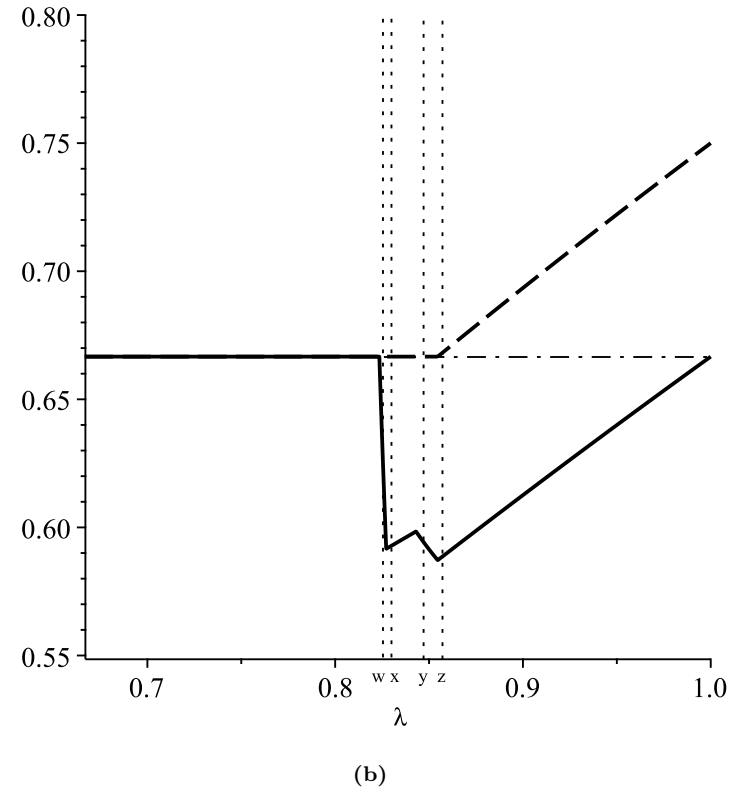
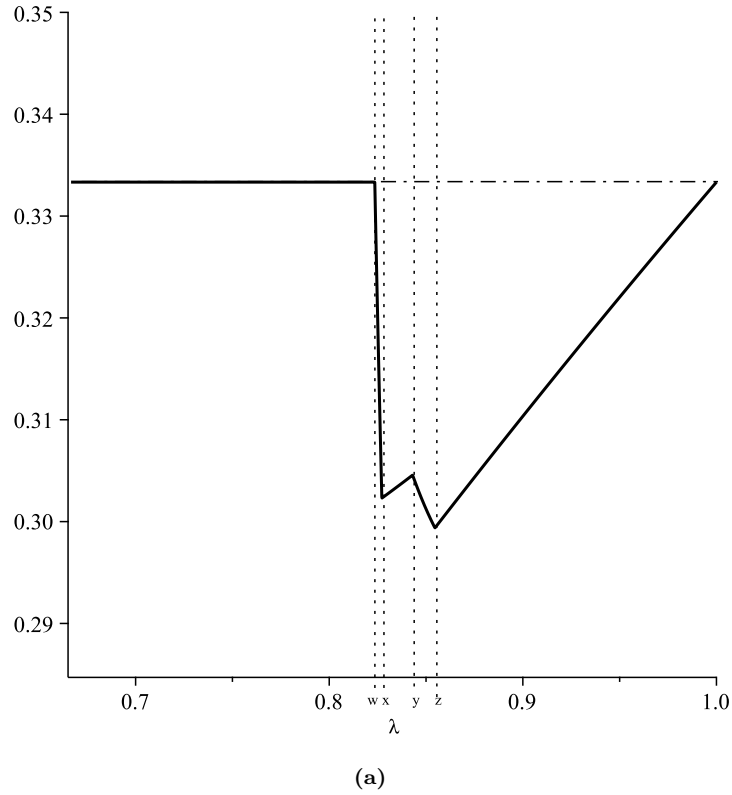
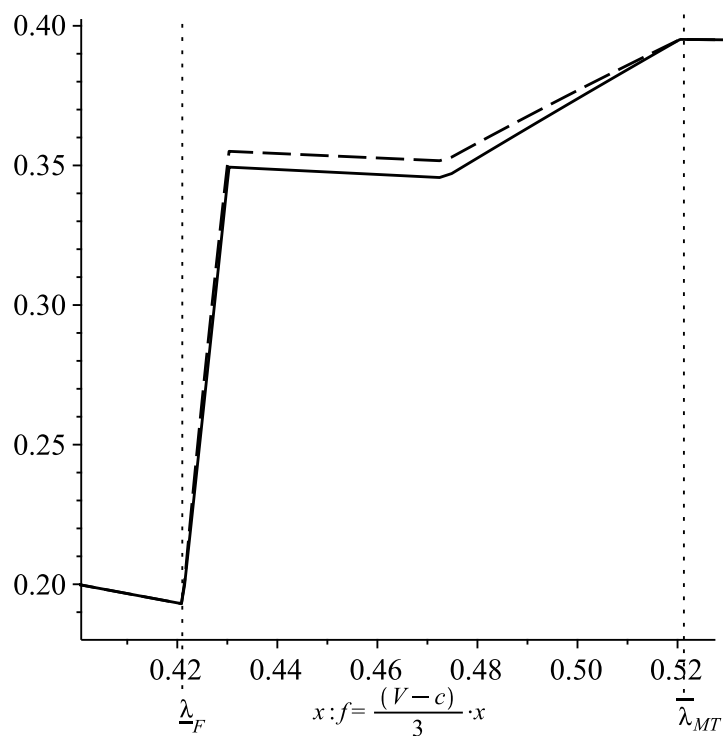
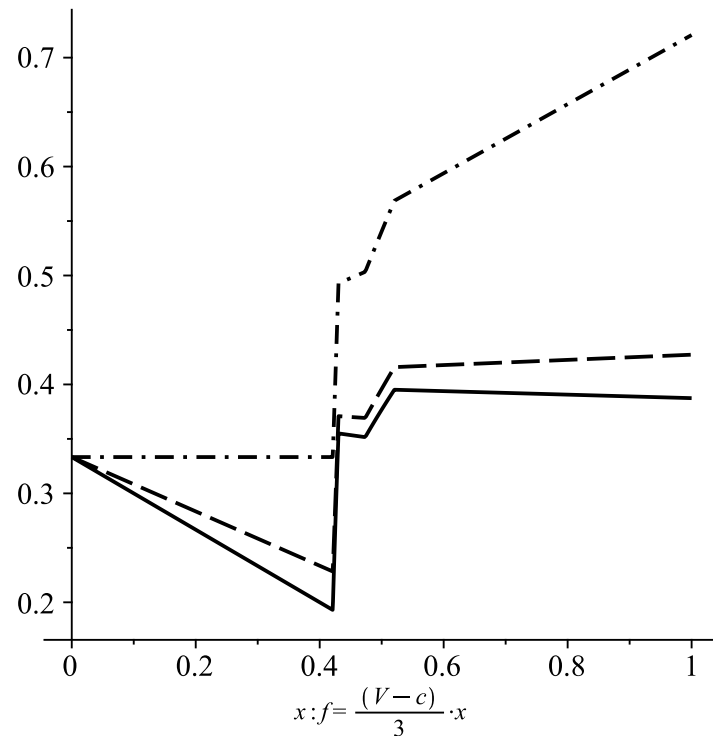


Figure 3: Bid-Ask Spreads and the Fee Split

Panel (a) below plots the quoted bid-ask spread S (dashed line) and the effective spread (solid line) in the value of the fee split, x , where x is such that $f = \frac{(V-c)x}{3}$. Panel (b) plots 3 different effective spreads in x : i) the raw effective spread E (solid line), ii) the average effective cum-fee spread, ACF (dashed line), and iii) the empirical effective cum-fee spread, ECF (dash-dot line). Parameters $V = 1$, $c = 0$, and $\lambda = 0.85$. Results for other values of V , c , and λ are qualitatively similar.



(a)



(b)

B Online Appendix

Proof (Proposition 5).

To show that $q_0(\lambda, x)$ is positive for all $\lambda \in [\underline{\lambda}_F, \bar{\lambda}_F]$, and $q_1(\lambda, x)$ is positive for all $\lambda \in [\underline{\lambda}_{MT}, \bar{\lambda}_{MT}]$, we plot them for all $\lambda \in [\underline{\lambda}_F, 1]$ and $x \in [0, 1]$. Writing out $q_0(x, \lambda)$ and $q_1(x, \lambda)$ explicitly.

$$q_0(\lambda, x) = -32 + 16(2 + x) + 8(29 - 10x - x^2)\lambda^2 - 4(73 + 8x - 3x^2)\lambda^3 \quad (95)$$

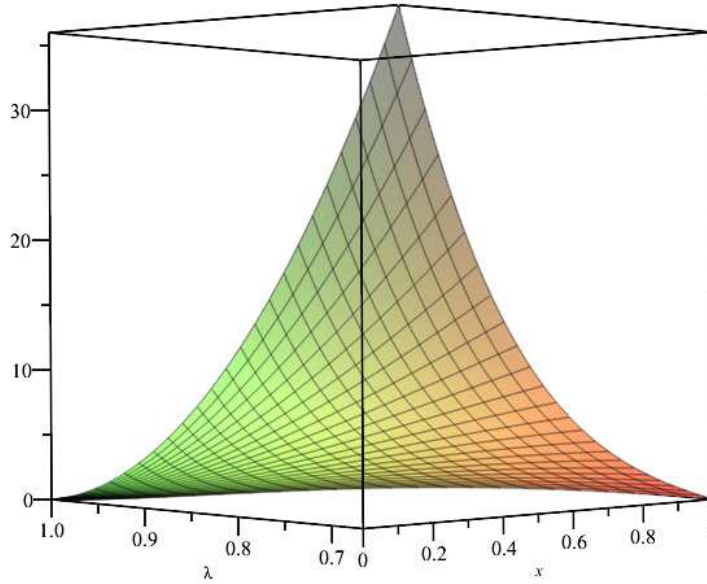
$$+ 6(4 + 6x + 3x^2)\lambda^4 + (16 + 20x + 9x^2)\lambda^5 + 4(8 + 3x + x^2)\lambda^6 + (2 + x)^2\lambda^7$$

$$q_1(\lambda, x) = 352 + (976 + 384x)\lambda - (936 + 992x - 128x^2)\lambda^2 - (596 + 368x - 128x^2)\lambda^3 \\ - (788 + 280x - 408x^2)\lambda^4 - (292 + 868x + 644x^2)\lambda^5 \quad (96)$$

$$+ (484 + 676x + 470x^2)\lambda^6 + (200 + 152x + 171x^2)\lambda^7 + (64 - 152x - 9x^2)\lambda^8$$

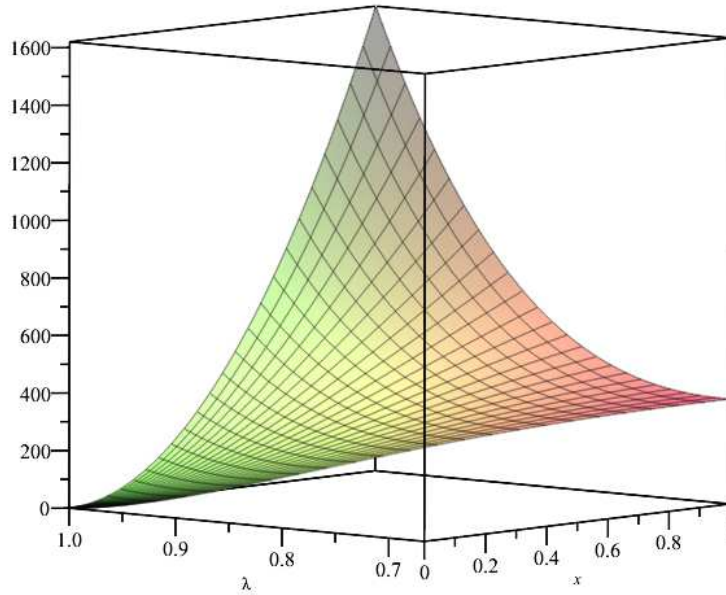
$$- (44 + 208x + 45x^2)\lambda^9 - (4 + 64x - 15x^2)\lambda^{10} + (4 - 12x - 3x^2)\lambda^{11} - (2 + x)^2\lambda^{12}$$

Plotting $q_0(\lambda, x)$ from equation (95) yields:



By inspection, the figure is non-negative for all $\lambda \in [\underline{\lambda}_F, 1]$ and $x \in [0, 1]$. Because the function converges to its lowest points at $(\lambda, x) = \{(1, 0), (\underline{\lambda}_F, 1)\}$, we evaluate the function at these points, finding that $q_0(1, 0) = q_0(\underline{\lambda}_F, 1) = 0$. We achieve the solution at $q_0(\underline{\lambda}_F, 1)$ by recalling that $\underline{\lambda}_F = \frac{2(4-x)}{8+x}$, which implies that $\underline{\lambda}_F(x = 1) = 2/3$.

Similarly, plotting $q_1(\lambda, x)$ in λ , we have:

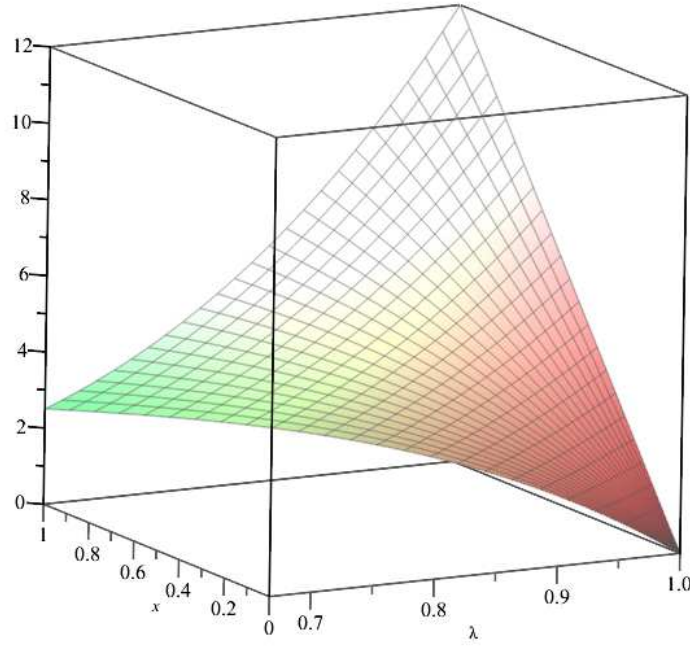


By inspection, the figure is above zero for all $\lambda \in [\underline{\lambda}_F, 1)$ and $x = (0, 1]$. Because the function converges to its lowest point at $(\lambda, x) = (1, 0)$, we evaluate the function at this point, finding that equals $q_1(1, 0) = 0$.

Next, we show that $r_0(\lambda, x)$ is positive for all $\lambda \in [\underline{\lambda}_F, \bar{\lambda}_F]$, and $r_1(\lambda, x)$ is positive for all $\lambda \in [\underline{\lambda}_{MT}, \bar{\lambda}_{MT}]$, by plotting them for all $\lambda \in [\underline{\lambda}_F, 1]$ and $x = [0, 1]$. First, we write r_0 explicitly.

$$r_0 = 4 - 12\lambda + 15\lambda^2 - (23 - 8x)\lambda^3 + (21 + 4x)\lambda^4 - 5\lambda^5. \quad (97)$$

Plotting $r_0(\lambda, x)$ from equation (97) yields:

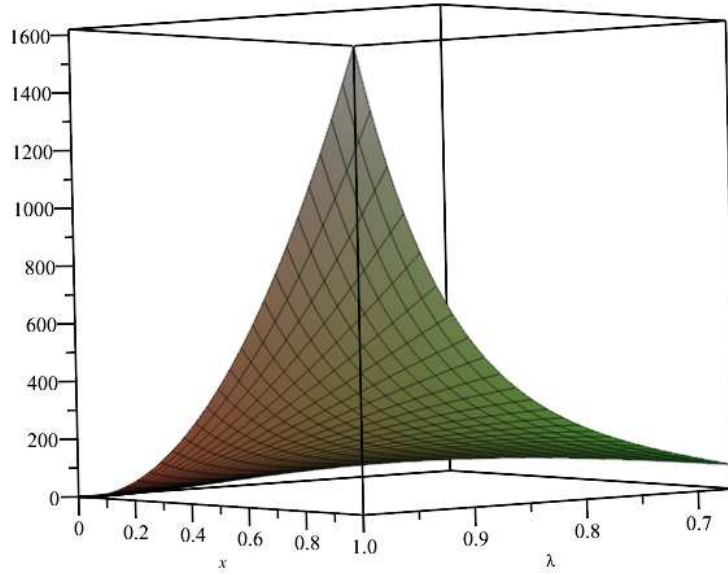


By inspection, the figure is above zero for all $\lambda \in [\underline{\lambda}_F, 1)$ and $x = (0, 1]$. Because the function converges to its lowest point at $(\lambda, x) = (1, 0)$, we evaluate the function at this point, finding that equals $r_0(1, 0) = 0$.

Next, we write $r_1(\lambda, x)$ below.

$$\begin{aligned}
 r_1(\lambda, x) = & -128 - 64\lambda + 128(1+x)\lambda^2 - 32(3+2x)\lambda^3 + (440 + 384x - 32x^2)\lambda^4 \\
 & + (1100 + 608x + 48x^2)\lambda^5 - (484 + 568x + 128x^2)\lambda^6 - (444 + 884x - 56x^2)\lambda^7 \\
 & - (1212 + 1036x - 198x^2)\lambda^8 - (56 - 152x + 515x^2)\lambda^9 + (224 + 608x + 531x^2)\lambda^{10} \\
 & + (356 + 504x + 327x^2)\lambda^{11} + (188 + 192x + 117x^2)\lambda^{12} + (36 - 12x - 3x^2)\lambda^{13} \\
 & + 3(4(1-x) - 3x^3)\lambda^{14}
 \end{aligned} \tag{98}$$

Plotting $r_1(\lambda, x)$ in λ , we have:



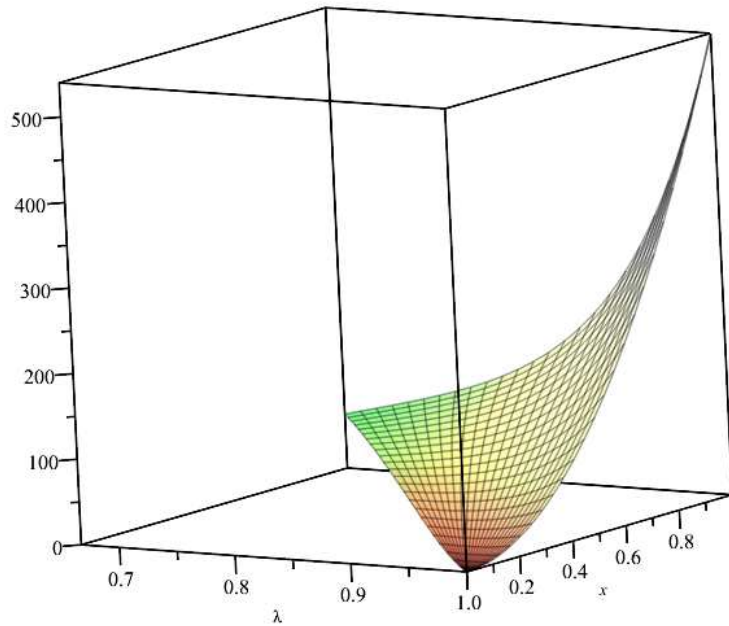
By inspection, the figure is above zero for all $\lambda \in [\underline{\lambda}_F, 1)$ and $x = (0, 1]$. Because the function converges to its lowest point at $(\lambda, x) = (1, 0)$, we evaluate the function at this point, finding that equals $r_1(1, 0) = 0$. $\square$

Proof (Proposition 6). To show that $r_2(\lambda, x)$ is positive for all $\lambda \in [\underline{\lambda}_{MT}, \bar{\lambda}_{MT}]$, we

plot the function for all $\lambda \in [\lambda_F, 1]$ and $x = [0, 1]$. We begin by writing $r_2(\lambda, x)$ below.

$$\begin{aligned}
 r_2(\lambda, x) = & -256 - 192\lambda + (608 + 512x)\lambda^2 + (1040 + 448x)\lambda^3 - (264 + 416x + 128x^2)\lambda^4 \\
 & - (296 + 720x - 16x^2)\lambda^5 - (1080 + 888x - 152x^2)\lambda^6 - (136 + 40x - 436x^2)\lambda^7 \\
 & - (32 - 304x - 474x^2)\lambda^8 + (240 + 400x + 362x^2)\lambda^9 + (200 + 256x + 202x^2)\lambda^{10} \\
 & + (104 + 96x + 74x^2)\lambda^{11} + (56 + 40x + 26x^2)\lambda^{12} + 2(4(1+x) + 3x^2)\lambda^{13}
 \end{aligned} \tag{99}$$

Plotting $r_2(\lambda, x)$, we have:



By inspection, the figure is above zero for all $\lambda \in [\lambda_F, 1)$ and $x = (0, 1]$. Because the function converges to its lowest point at $(\lambda, x) = (1, 0)$, we evaluate the function at this point, finding that equals $r_2(1, 0) = 0$. $\square$